Informatique et systèmes de communication ISC
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

Data Processing and Visualization

C3. Tabular 2
Introduction to Data Visualization

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Date

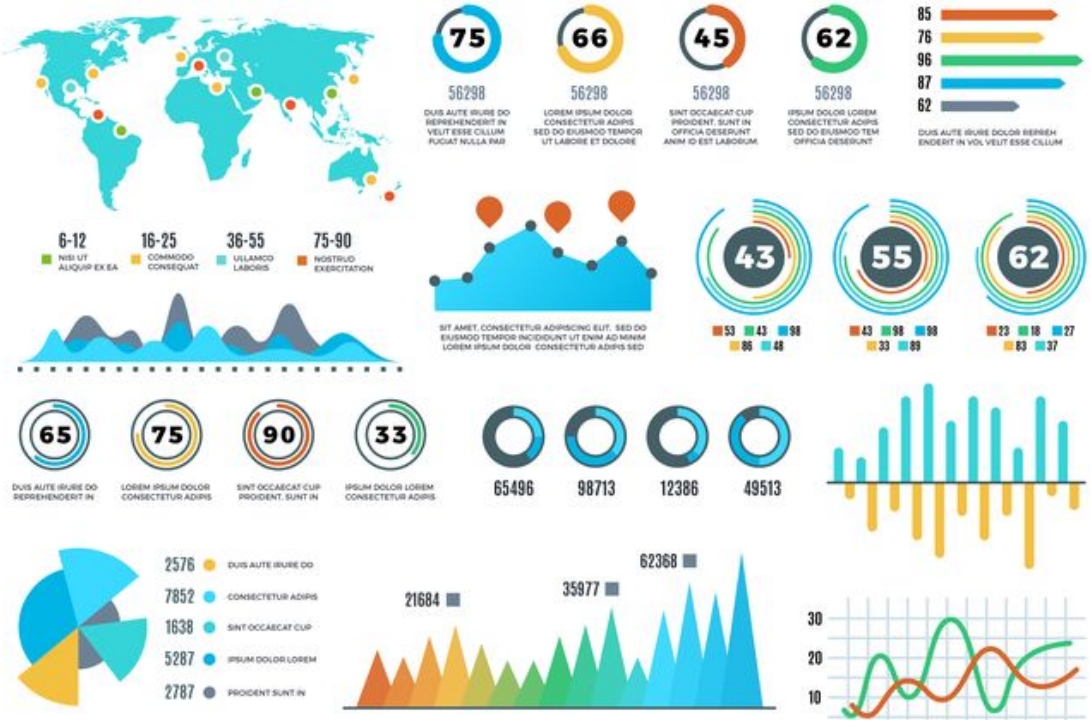
Background

Background

- Why data visualization ?
  - Helps to tell stories
  - Present complex, large amount of data, in a form easier to understand
  - Highlight the trends
  - Back up your claims
  - Illustrate your point

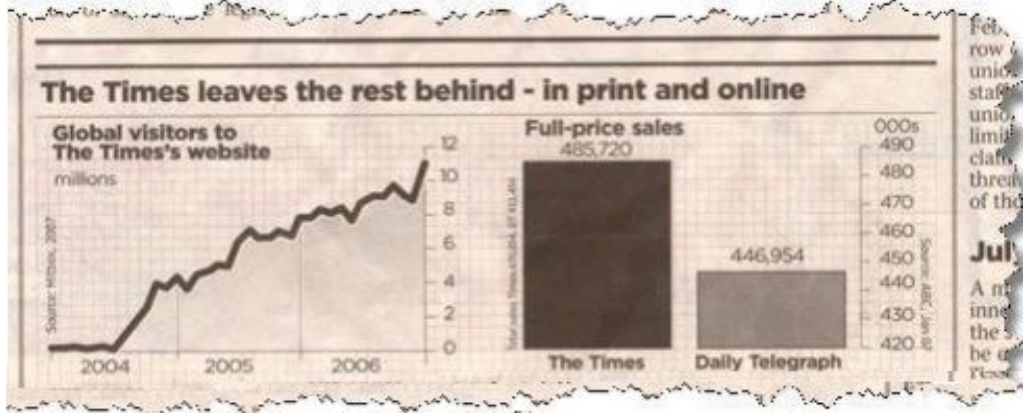
			Bright	DateTime	Time	ddist	dh	dist	dpos	dv	h	lat	lon	v	x	y
Night	Station	Event														
0	0	0	3.3	2011-12-13 22:15:37.880	37.88	----.-	----.-	115747.2	6.0	--.---	99720.4	28.532	-17.091	--.---	0.547	0.633
		1	3.1	2011-12-13 22:15:37.920	37.92	----.-	----.-	115629.8	6.1	--.---	99037.5	28.527	-17.103	35.203	0.536	0.607
		2	2.5	2011-12-13 22:15:37.970	37.97	----.-	----.-	115528.0	3.8	--.---	98328.5	28.523	-17.115	29.251	0.524	0.579
		3	2.2	2011-12-13 22:15:38.010	38.01	----.-	----.-	115447.1	2.6	--.---	97654.7	28.518	-17.126	34.76	0.514	0.552
		4	2.1	2011-12-13 22:15:38.040	38.04	----.-	----.-	115378.7	1.4	--.---	96957.0	28.514	-17.138	48.01	0.503	0.525
		5	1.8	2011-12-13 22:15:38.080	38.08	----.-	----.-	115333.0	3.4	--.---	96292.1	28.509	-17.149	34.324	0.492	0.499
		6	1.7	2011-12-13 22:15:38.130	38.13	----.-	----.-	115304.3	3.4	--.---	95580.9	28.504	-17.161	29.381	0.481	0.471
		7	1.4	2011-12-13 22:15:38.170	38.17	----.-	----.-	115292.9	5.4	--.---	94905.6	28.500	-17.172	34.882	0.470	0.444
		8	1.2	2011-12-13 22:15:38.210	38.21	----.-	----.-	115298.7	2.1	--.---	94233.5	28.495	-17.184	34.733	0.459	0.417
		9	1.0	2011-12-13 22:15:38.240	38.24	----.-	----.-	115319.8	4.4	--.---	93540.6	28.491	-17.195	47.763	0.448	0.390
		10	1.0	2011-12-13 22:15:38.280	38.28	----.-	----.-	115359.5	6.4	--.---	92880.9	28.487	-17.206	34.114	0.438	0.364
		11	1.5	2011-12-13 22:15:38.330	38.33	----.-	----.-	115417.3	7.8	--.---	92230.9	28.482	-17.217	26.898	0.427	0.339
		12	2.0	2011-12-13 22:15:38.370	38.37	----.-	----.-	115487.6	10.8	--.---	91590.6	28.478	-17.228	33.134	0.417	0.313
		13	2.5	2011-12-13 22:15:38.400	38.40	----.-	----.-	115568.6	1.6	--.---	90968.2	28.474	-17.239	42.954	0.407	0.289
		14	3.0	2011-12-13 22:15:38.440	38.44	----.-	----.-	115673.7	5.4	--.---	90299.8	28.469	-17.250	34.609	0.397	0.263
		15	2.8	2011-12-13 22:15:38.490	38.49	----.-	----.-	115784.5	6.3	--.---	89708.9	28.465	-17.260	24.487	0.388	0.239
		16	4.0	2011-12-13 22:15:38.530	38.53	----.-	----.-	115908.0	5.5	--.---	89106.8	28.461	-17.270	31.197	0.378	0.216
		17	3.4	2011-12-13 22:15:38.560	38.56	----.-	----.-	116061.4	10.3	--.---	88420.0	28.457	-17.282	47.462	0.367	0.189
		18	4.4	2011-12-13 22:15:38.600	38.60	----.-	----.-	116180.4	3.9	--.---	87955.7	28.454	-17.290	24.072	0.360	0.171
		0	4.5	2011-12-13 22:15:37.970	37.97			127788.5	0.6		99970.0	28.533	-17.087		0.446	0.282

VS



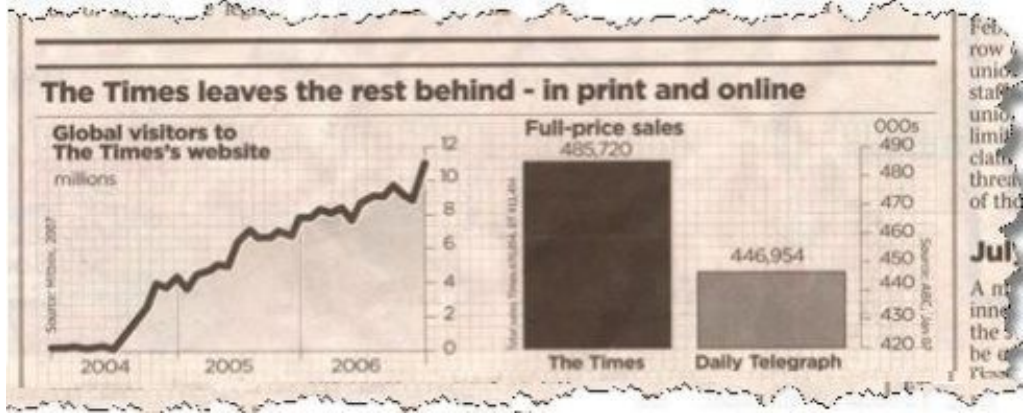
neilpatel.com/blog/data-visualization

Background



- Why data visualization ?
  - Humans are used to accepting graphics uncritically, not asking enough questions of them
  - Questions + Criticism of
    - Others' graphs
    - Your own graphs
    - Conclusions drawn (can we generalize? Bias in the data? Correlation is not causation)

Background



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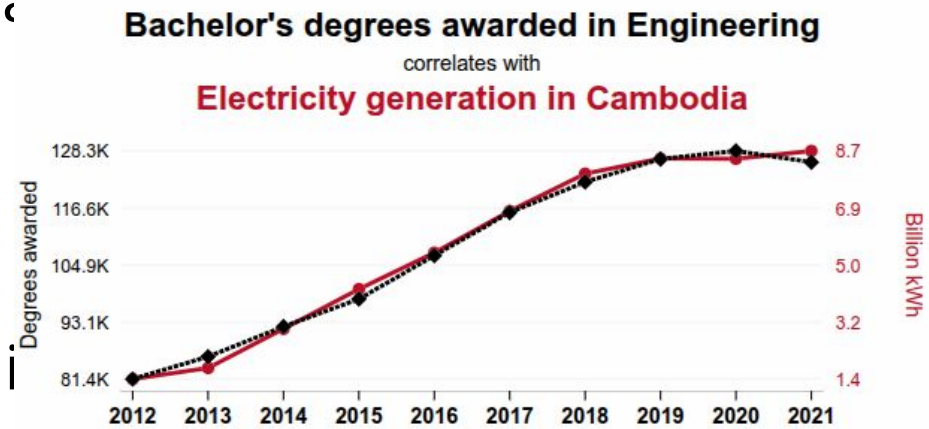
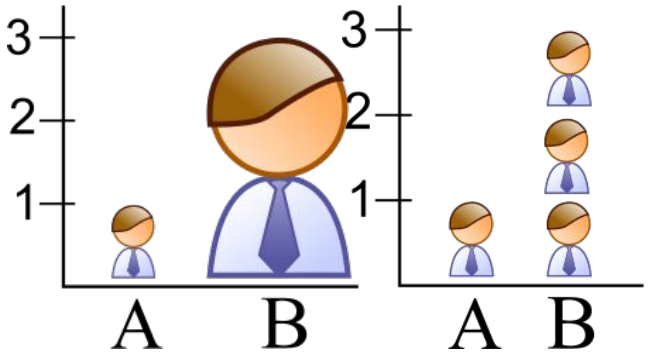
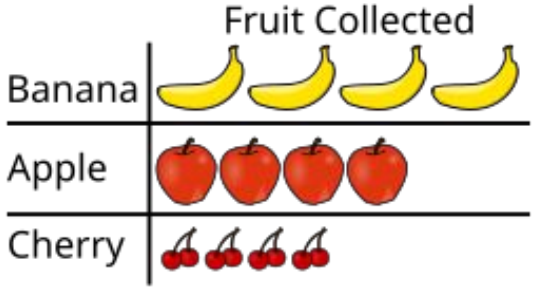
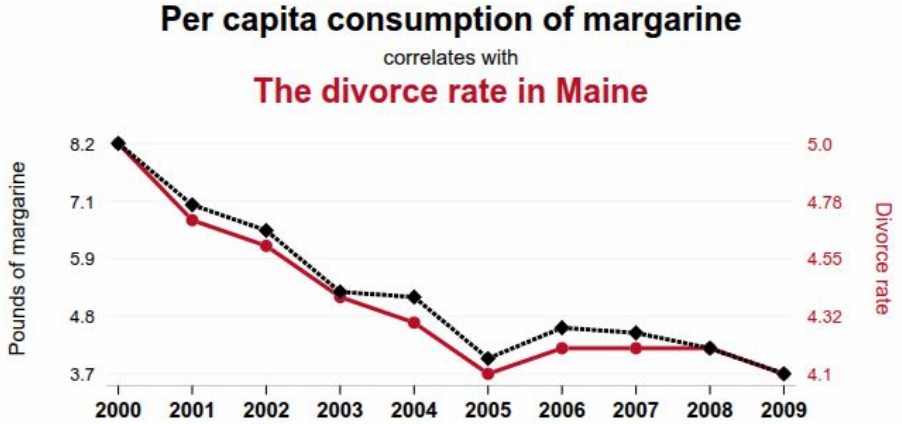








1. Principles and Guidelines for Effective Data Visualization

1. Principles and Guidelines for Effective Data Visualization

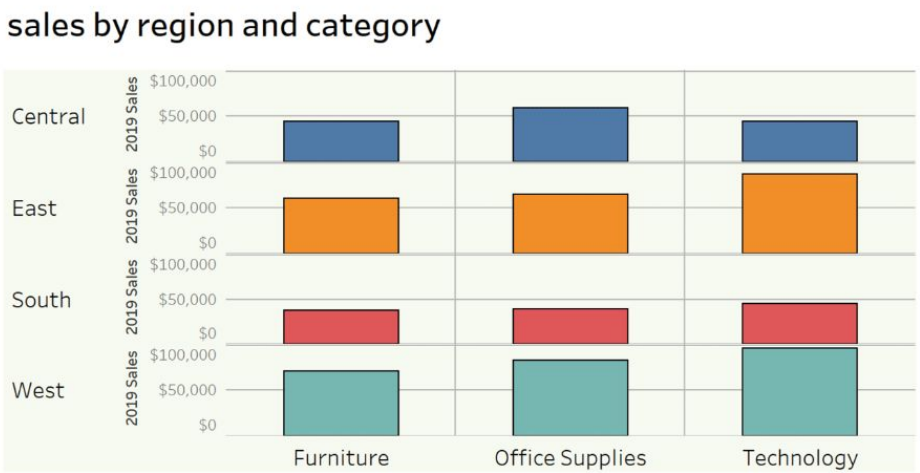
- Effective Data Visualization
  - Accuracy: Check the data sources, values, be critical with results/conclusions
  - Clarity: Label axes, clear title to avoid confusion
  - Text or explanations that go with it (data, trends, conclusions etc.)
  - Audience considerations: should guide exploration and presentation
  - Color theory and accessibility: Consider Colour Vision Deficiency (CVD)
    - high-contrast color combinations, avoid color-coding data, and provide text descriptions
    - blue/orange palette instead of red/green
    - Palettes: ‘viridis’, ‘cividis’, ‘colorblind’
  - Simple visualization, less is more

1. Principles and Guidelines for Effective Data Visualization

- Common Mistakes and Pitfalls
  - Overcomplicating visuals
  - Misleading representations
  - Choosing inappropriate chart types

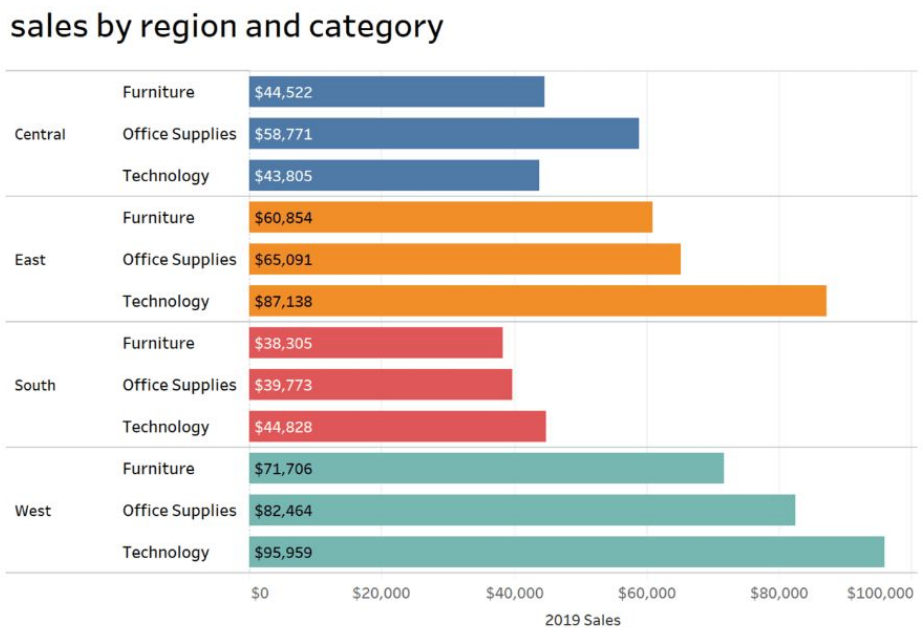
1. Principles and Guidelines for Effective Data Visualization

- Common Mistakes and Pitfalls
  - Overcomplicating visuals
  - M
  - C



	Central	East	South	West
Furniture	$44,522	$60,854	$38,305	$71,706
Office Supplies	$58,771	$65,091	$39,773	$82,464
Technology	$43,805	$87,138	$44,828	$95,959

ineffective



effective

1. Principles and Guidelines for Effective Data Visualization

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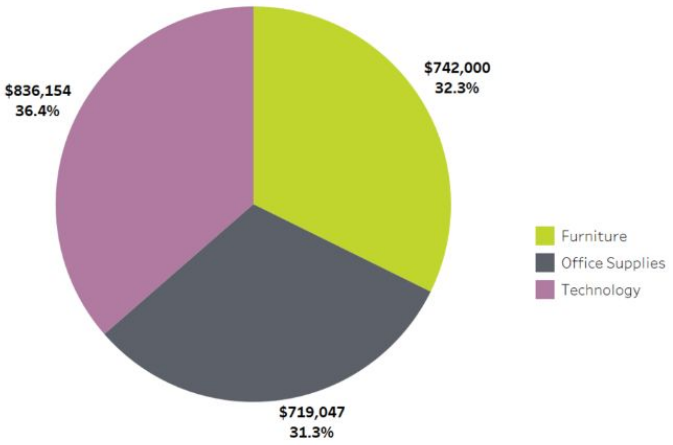
sales by product category



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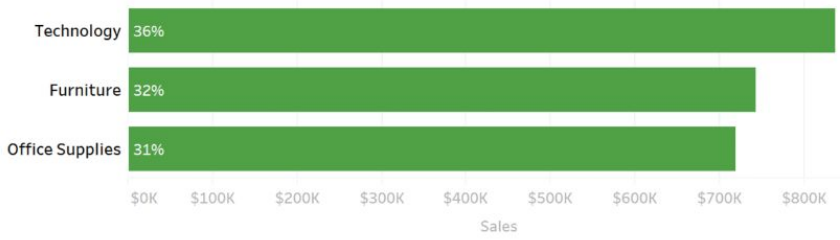
sales by product category



effective

1. Principles and Guidelines for Effective Data Visualization

- Common Mistakes and Pitfalls
  - Overcomplicating visuals
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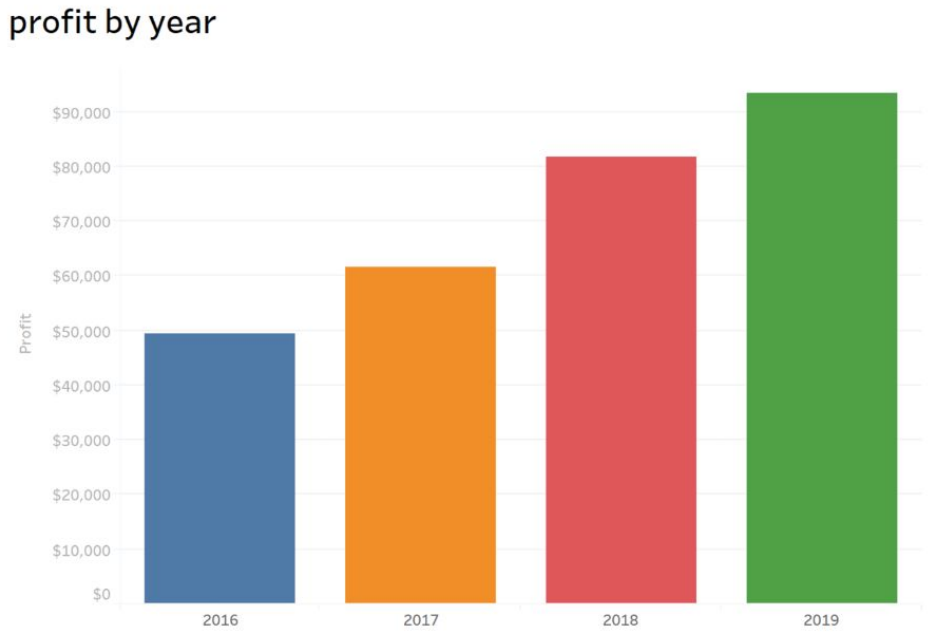
  - Ch



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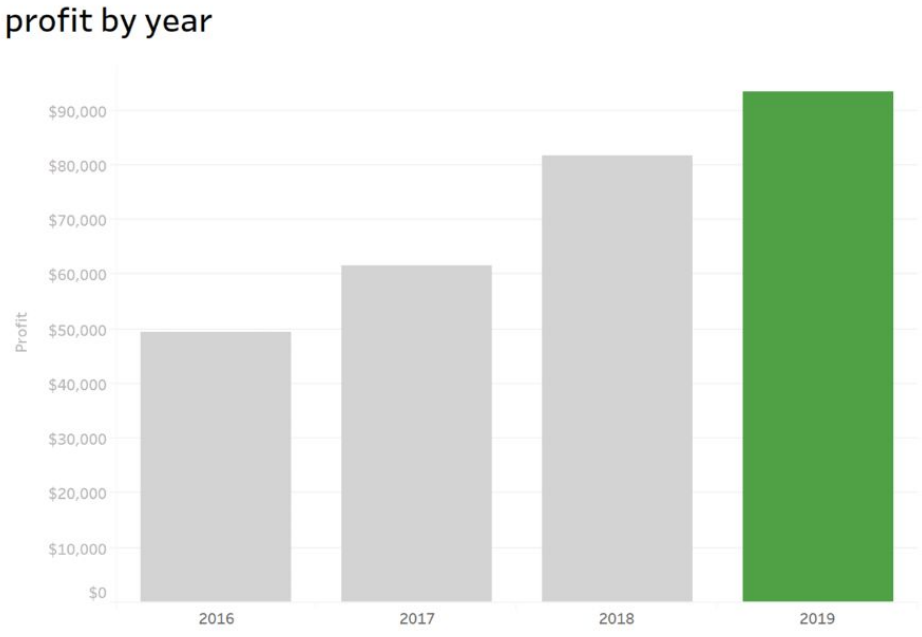


effective

2. Data Visualization Tools and Libraries

2. Data Visualization Tools and Libraries

- Popular tools:
  - **Python libraries: Matplotlib, plotly, seaborn, etc.**
  - Excel
  - D3.js (JavaScript library),
  - PowerBI (Microsoft)
  - Many more, some free, some not

2. Data Visualization Tools and Libraries

Python libraries:

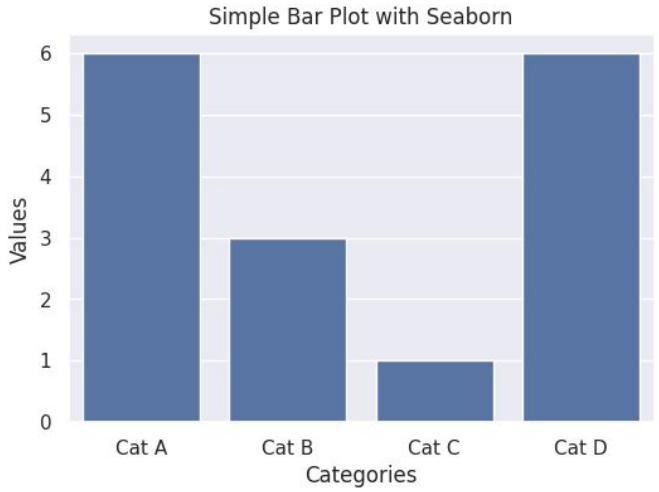
- NumPy wide variety of mathematical operations on arrays. It adds powerful data structures to Python that guarantee efficient calculations with arrays and matrices and it supplies many high-level mathematical functions that operate on these arrays and matrices
- **Matplotlib** library for creating static, animated, and interactive visualizations
  - **Pandas** is built on top of *matplotlib* for data visualization and *NumPy* for mathematical operations. It acts as a wrapper over these libraries
  - **Seaborn** is a data visualization library based on *matplotlib*
- **Plotly** is a library for data visualization. It produces interactive graphs and can be embedded on websites

3. Creating Basic Visualizations

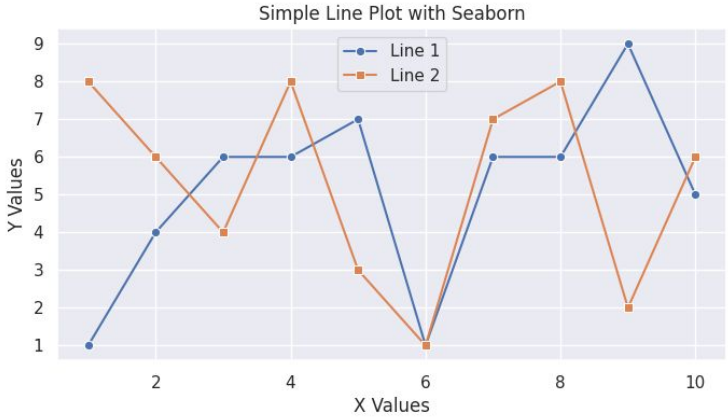
3. Creating Basic Visualizations

- Bar Charts and Line Plots
- Crosstab and Stacked bar charts
- Scatter Plots, Histograms, probability density

3. Creating Basic Visualizations



- **Bar Charts and Line Plots**
  - Bar chart:
    - Compare things between different groups
    - Track changes over time (big changes)
  - Line plot:
    - Track changes over time (small and big changes)
    - Compare changes over the same period of time for more than one group

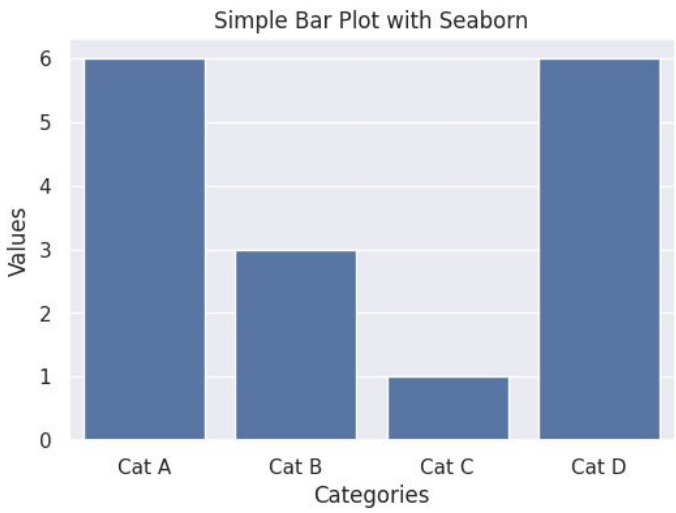


3. Creating Basic Visualizations

- Bar charts example

```
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

# Activating seaborn's "default" theme
sns.set_theme()

# Generating data
categories = ['Cat A', 'Cat B', 'Cat C', 'Cat D']
values = np.random.randint(1, 10, size=len(categories))

# Plotting the bar plot
plt.figure(figsize=(6, 4))
sns.barplot(x=categories, y=values)

# Adding labels and title
plt.xlabel("Categories")
plt.ylabel("Values")
plt.title("Simple Bar Plot with Seaborn")

# Display the plot
plt.show()
```

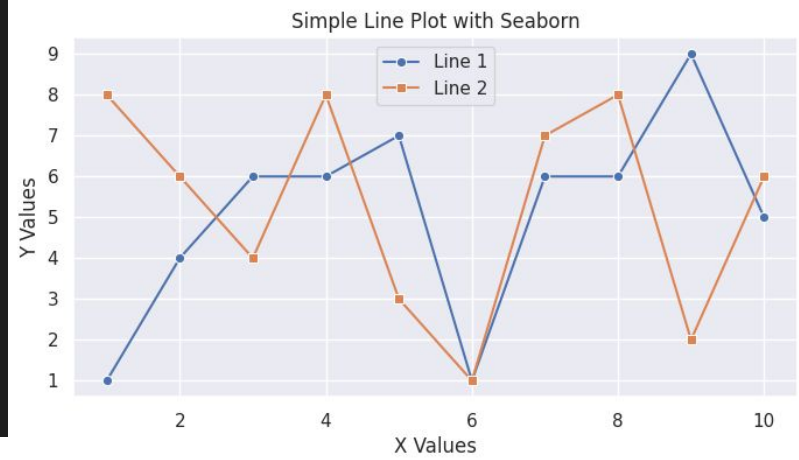
```
print(values)
✓ 0.0s
[6 3 7 2]
```



3. Creating Basic Visualizations

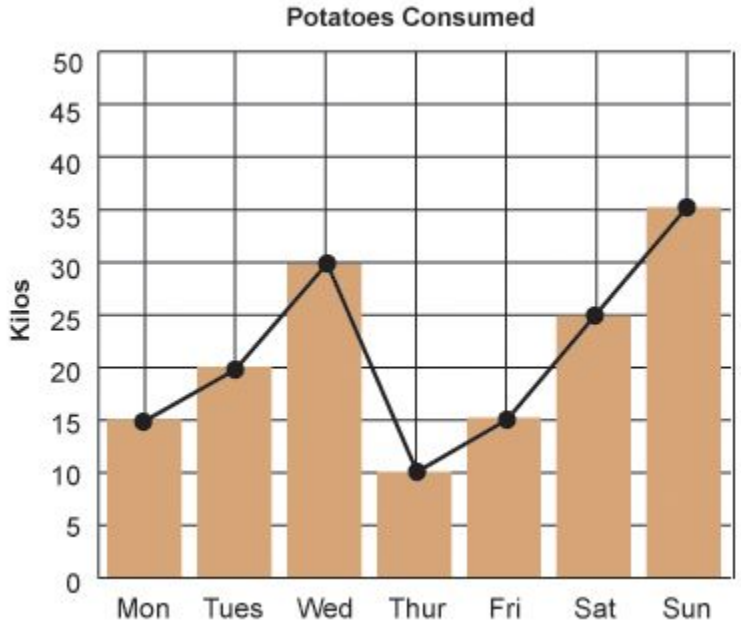
- Line plot example

```
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
# Generating data
x_values = np.arange(1, 11)
y_values1 = np.random.randint(1, 10, size=len(x_values))
y_values2 = np.random.randint(1, 10, size=len(x_values))

# Plotting the line plot
plt.figure(figsize=(8, 4))
sns.lineplot(x=x_values, y=y_values1, label='Line 1', marker='o')
sns.lineplot(x=x_values, y=y_values2, label='Line 2', marker='s')

# Adding labels and title
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.title("Simple Line Plot with Seaborn")

# Display the legend
plt.legend()
# Display the plot
plt.show()
```

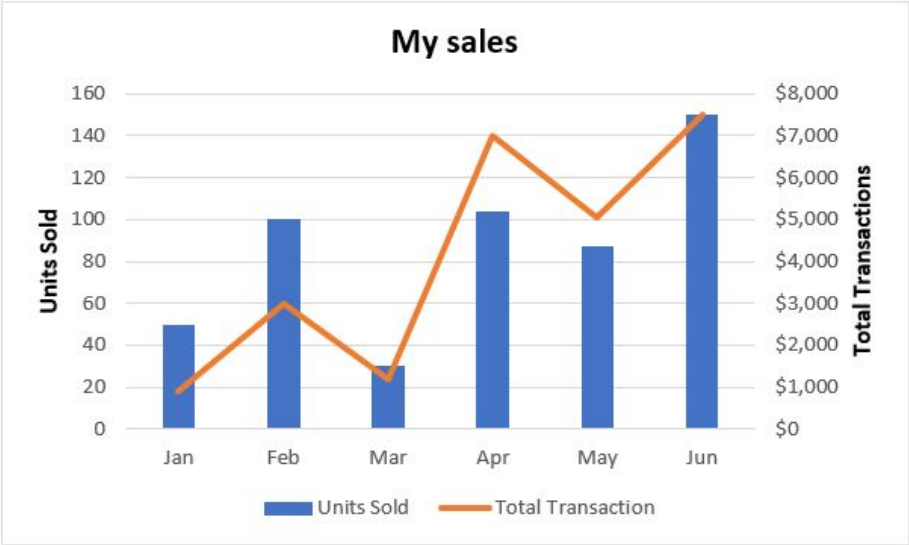


3. Creating Basic Visualizations

- Bar Charts and Line Plots
  - Can be combined



Same variable

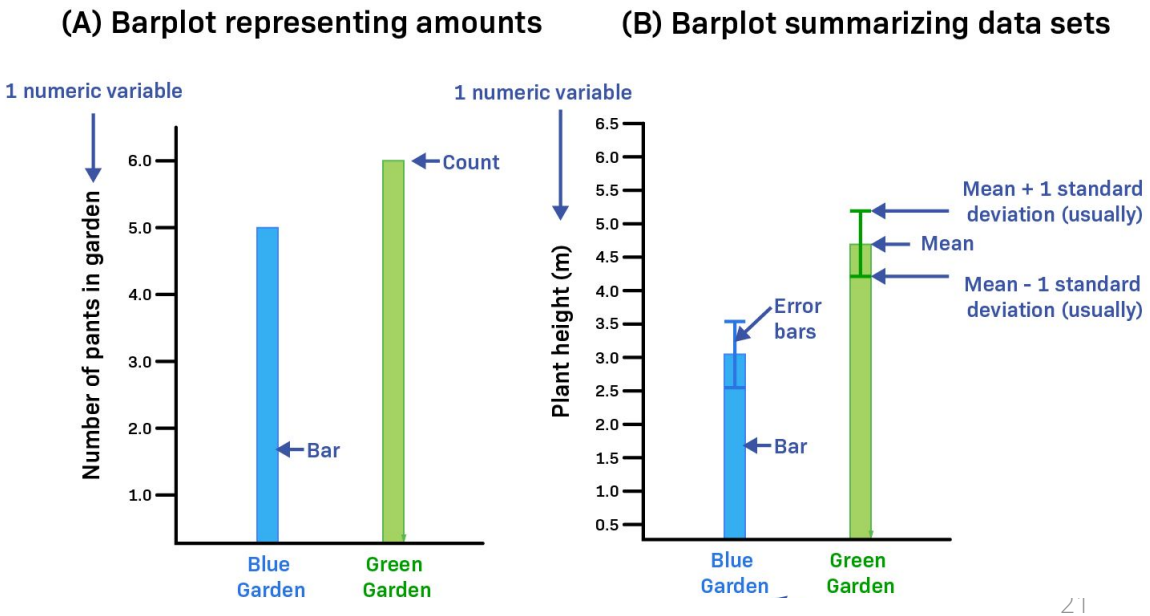


Different variables

3. Creating Basic Visualizations

- Bar Charts and Line Plots
  - Error bars:
    - When the values in the plot are the **result of an aggregation** (e.g. mean)
    - Shows the variance of the values
      - [10.1, 10.2, 10.1, 9.9] low variance
      - [6.1, 12.3, 10.2, 13.8, 8.4] high variance
    - Usually +/- standard deviation

www.labxchange.org



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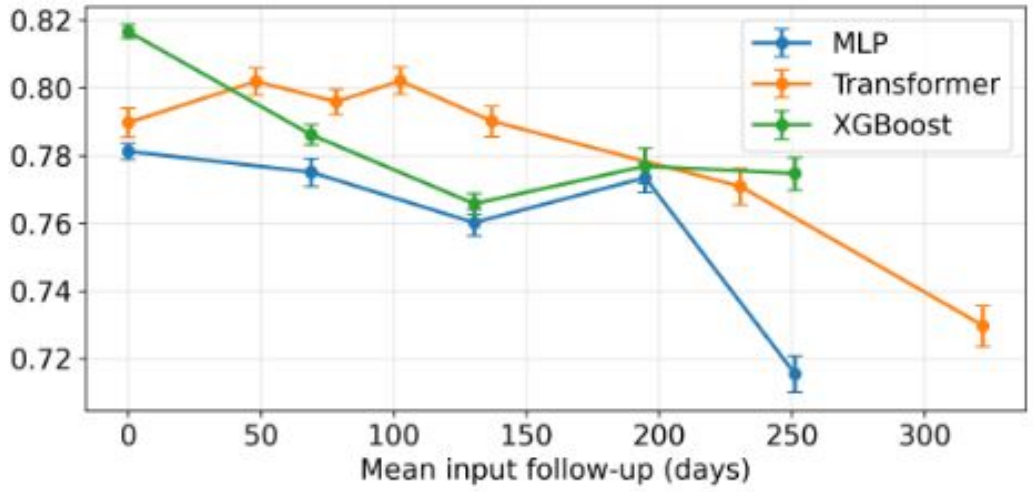
      - [illegible] variance



(d) AUCs horizon 15 months

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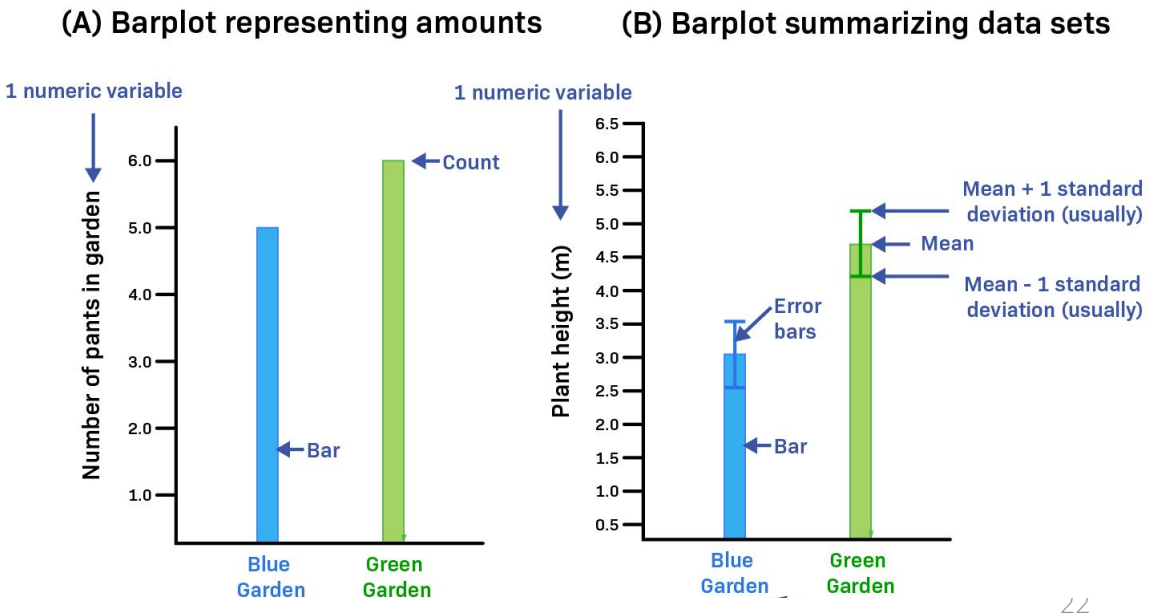


Exercises

3.1 Creating a bar chart and a line plot with sample data

We use the iris.csv dataset

a) Make a barplot representing, for each variety, the counts of sepal lengths greater than the overall mean sepal length

b) Make a bar plot of the mean sepal length by variety, with error bars (standard deviation)

3. Creating Basic Visualizations

- **Crosstab and Stacked Bar Chart**

3. Creating Basic Visualizations

- **Crosstab**
  - Shows the relationship between two or more variables
  - Similar to pivot tables (Tabular 1), but for categorical data, and computes only frequency/count
  - Can be plotted with grouped or stacked bar charts

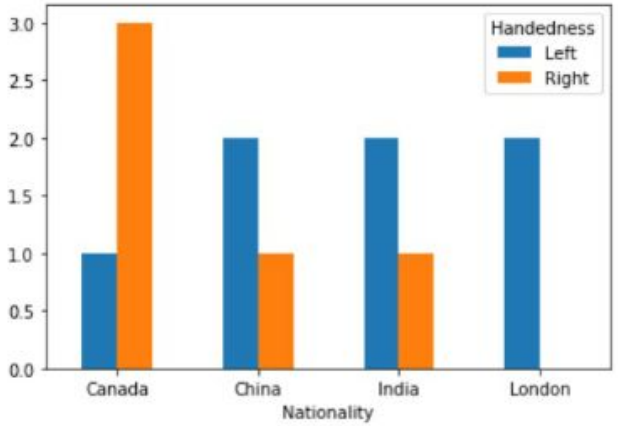
Name	Nationality	Sex	Age	Handedness
Kathy	Canada	Female	23	Right
nina	Canada	Female	18	Right
Peter	Canada	Male	19	Right
John	Canada	Male	22	Left
Fatima	London	Female	31	Left
Elizhabeth	London	Female	25	Left
Dhaval	India	Male	35	Left
Sudhir	India	Male	31	Left
Sanju	India	Male	18	Right
Yan	China	Female	52	Right
Juan	China	Female	58	Left
Liang	China	Male	43	Left

```
crosstb = pd.crosstab(index=df.Nationality, columns=df.Handedness)
```

Handedness	Left	Right
Nationality		
Canada	1	3
China	2	1
India	2	1
London	2	0

```
barplot = crosstb.plot.bar(rot=0)
```

grouped bar chart



https://www.geeksforgeeks.org/using-pandas-crosstab-to-create-a-bar-plot/

3. Creating Basic Visualizations

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Sanju	India	Male	18	Right
Yan	China	Female	52	Right
Juan	China	Female	58	Left
Liang	China	Male	43	Left

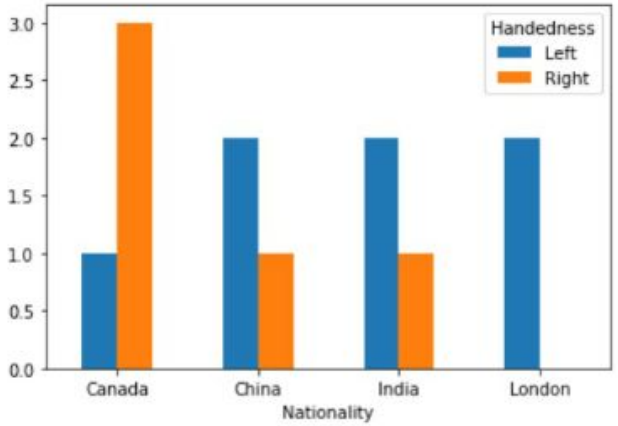
```
crosstb = df.pivot_table(index='Nationality', columns='Handedness', aggfunc=len)
```

```
crosstb = pd.crosstab(index=df.Nationality, columns=df.Handedness)
```

Handedness	Left	Right
Nationality		
Canada	1	3
China	2	1
India	2	1
London	2	0

```
barplot = crosstb.plot.bar(rot=0)
```

grouped bar chart



https://www.geeksforgeeks.org/using-pandas-crosstab-to-create-a-bar-plot/

3. Creating Basic Visualizations

- Crosstab
  - Note the difference: Presence of the “count” column

Name	Nationality	Sex	Age	Handedness
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Dhaval	India	Male	35	Left
Sudhir	India	Male	31	Left
Sanju	India	Male	18	Right
Yan	China	Female	52	Right
Juan	China	Female	58	Left
Liang	China	Male	43	Left

```
crosstb = pd.crosstab(df.Nationality, df.Handedness)
```

Handedness	Left	Right
Nationality		
Canada	1	3
China	2	1
India	2	1
London	2	0

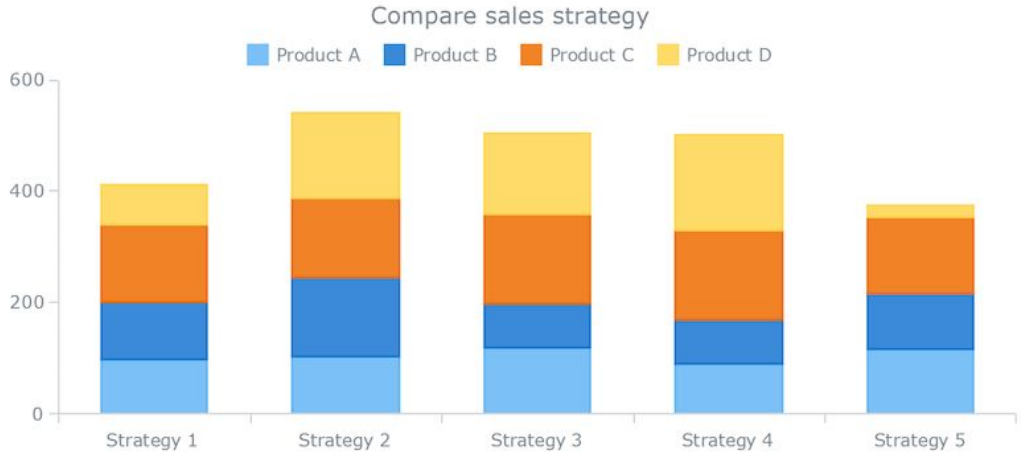
For the exercise

	nation	medal	count
0	South Korea	gold	24
1	China	gold	10
2	Canada	gold	9
3	South Korea	silver	13
4	China	silver	15

No need to count, use the “count” column instead.
df.pivot()
pd.crosstab()
pd.pivot_table()

3. Creating Basic Visualizations

- **Stacked Bar Chart**
  - Subgroups are displayed as bars on top of each other
    - Compare totals
    - Highlight sharp changes at the item level, that have most importance on movements in category totals

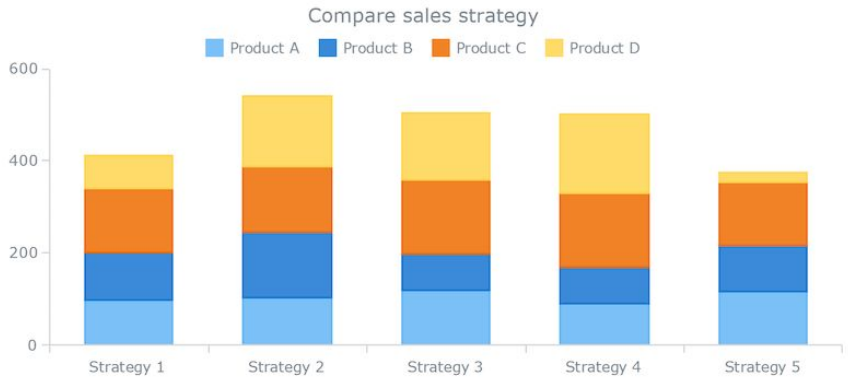


3. Creating Basic Visualizations

- Stacked Bar Chart
  - Subgroups are displayed as bars on top of each other
    - Compare totals
    - Highlight sharp changes at the item level, that have most importance on movements in category totals

Single-series bar chart
→ differences in category (strategy) totals

Multi-series/grouped bar chart
→ within strategy: products comparison
→ across strategies: product (e.g. A) comparison



Stacked bar chart
→ differences in category (strategy) totals
→ reasons for these differences (within/across strategies)

www.smashingmagazine.com/2017/03/understanding-stacked-bar-charts

	nation	medal	count
0	South Korea	gold	24
1	China	gold	10
2	Canada	gold	9
3	South Korea	silver	13
4	China	silver	15

Exercises

3.2 Building a Stacked Bar Chart and a Crosstab Chart

We use the medals_long.csv dataset: Olympic Short Track Speed Skating for the top three nations as of 2020.

a) Plot a stacked bar chart of medals counts per nation, stacking gold/silver/bronze. i.e. x: nations, y: stacked medal count (gold/silver/bronze)

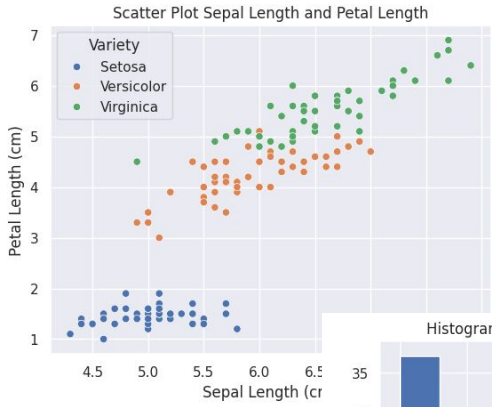
b) Plot a grouped bar chart of medals counts per nation

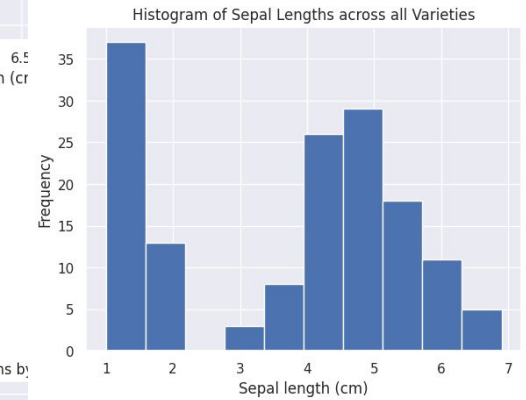
3. Creating Basic Visualizations

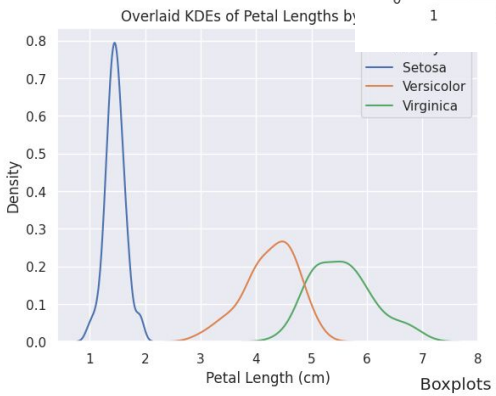
- Scatter Plots, Histograms, Box Plots, probability density

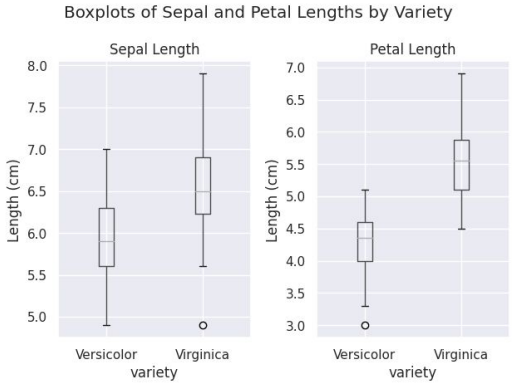
3. Creating Basic Visualizations

- Scatter Plot
  - Shows relationships between two numeric variables (positive or negative, non-linear or linear, strong or weak etc.)
- Histogram
  - Shows distribution of a feature
- Probability density
  - Estimates the probability density function of a continuous variable
- Box-Plot
  - Shows distribution of a feature (spread, skewness, median, outlier, etc.)

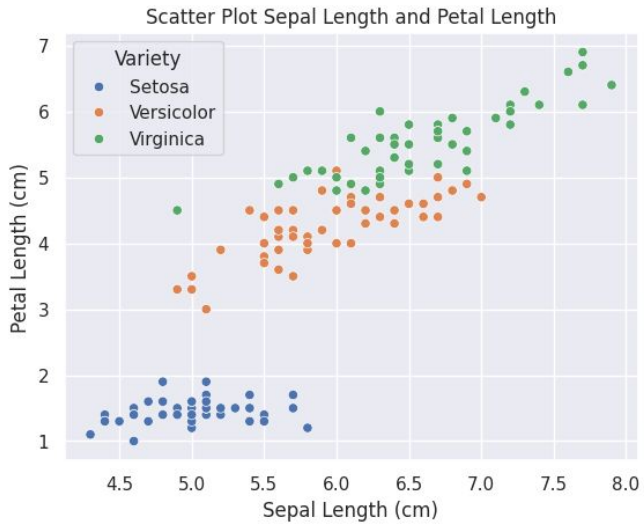




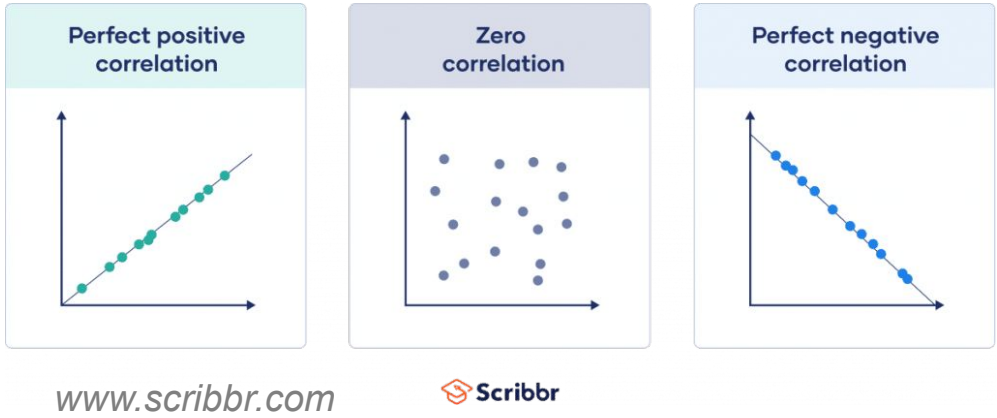


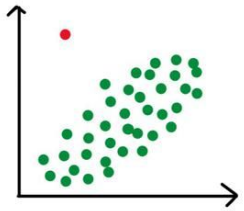


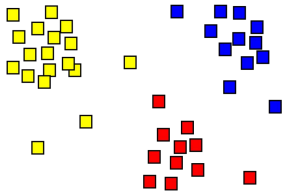
3. Creating Basic Visualizations



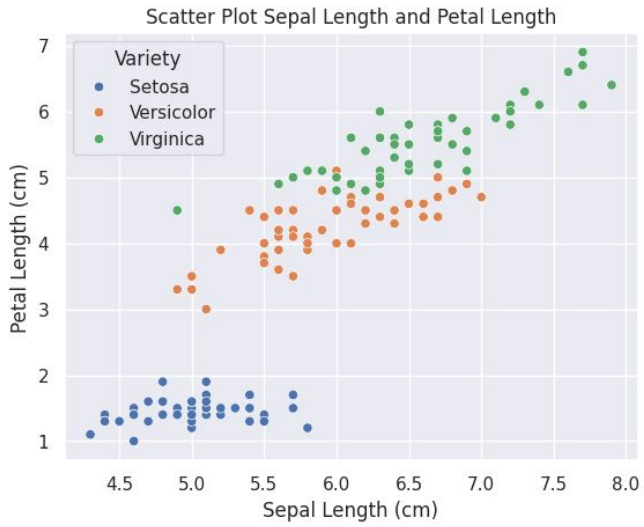
- **Scatter Plot**
  - Shows relationships between two numeric variables (positive or negative, non-linear or linear, strong or weak etc.)
  - Values of one variable on the x-axis
  - Values of the other variable on the y-axis
  - Each individual (observation) in the data appears as a point on the graph
  - Can reveal correlation, outliers, clusters







3. Creating Basic Visualizations



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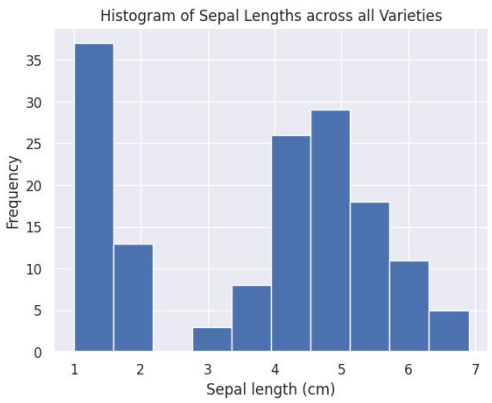
  - Can reveal correlation, outliers, clusters

sns.scatterplot()
df.plot.scatter()

www.scribbr.com

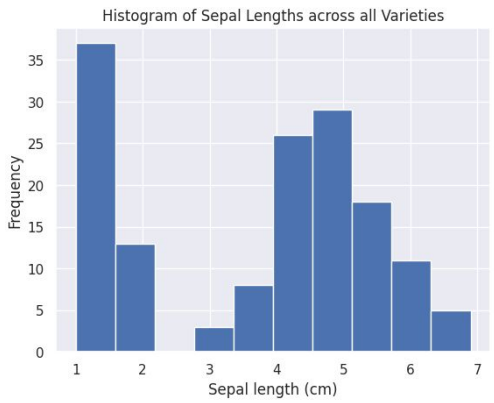
3. Creating Basic Visualizations



- **Histogram**
  - Shows distribution of a feature
  - Distribution: function that shows the possible values for a variable and how often they occur
  - 1. "Bin" the range of values (divide range into intervals)
  - 2. Count how many values fall into each interval
  - 3. Height of a rectangle represents the frequency (count) or relative frequency (count/total number)

sns.histplot()
df.plot.hist()
plt.hist()

3. Creating Basic Visualizations



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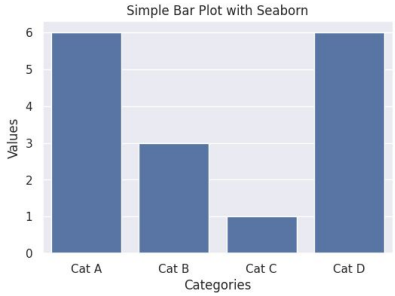
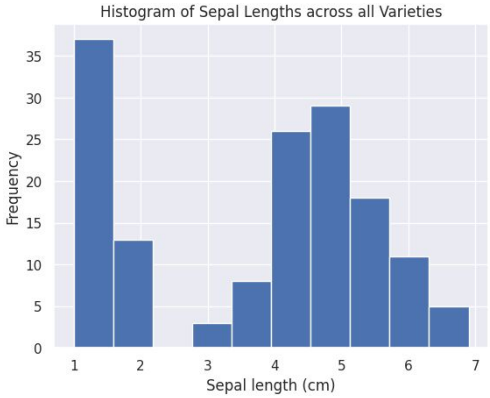
  - 2. Count how many values fall into each interval
  - 3. Height of a rectangle represents the frequency (count) or relative frequency (count/total number)

  - **Warning: not to confuse with bar chart !**
    Histogram : numerical data, the bins represent ranges of values
    Bar chart: plot of count of categorical variables
    (Notice the space between bars to avoid confusion)



3. Creating Basic Visualizations



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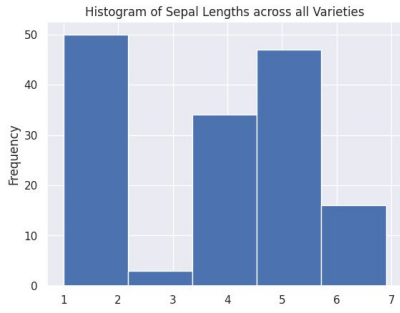
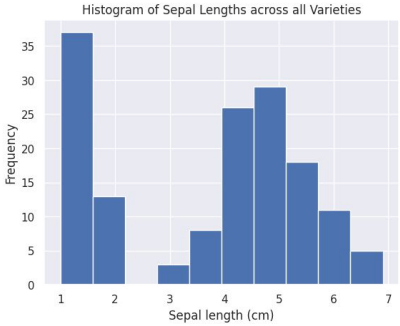
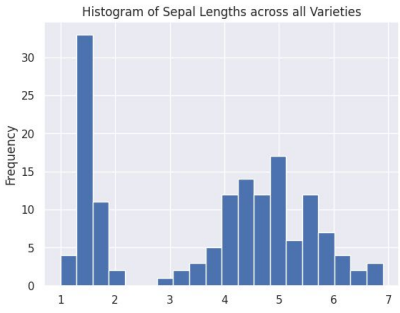
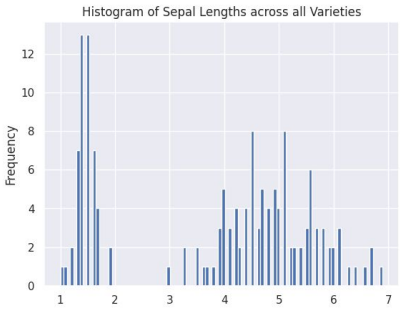
  - 2. Count how many values fall into each interval
  - 3. Height of a rectangle represents the frequency (count) or relative frequency (count/total number)

  - Problem: Depends on number of bins

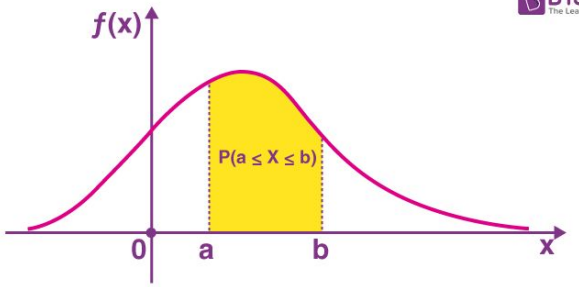






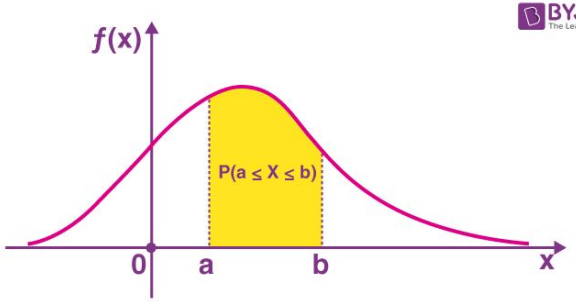


3. Creating Basic Visualizations



- **Probability density estimation**
  - Estimate, based on observed data, of an unobservable underlying probability density function

3. Creating Basic Visualizations



- Probability density estimation
  - Estimate, based on observed data, of an unobservable underlying probability density function

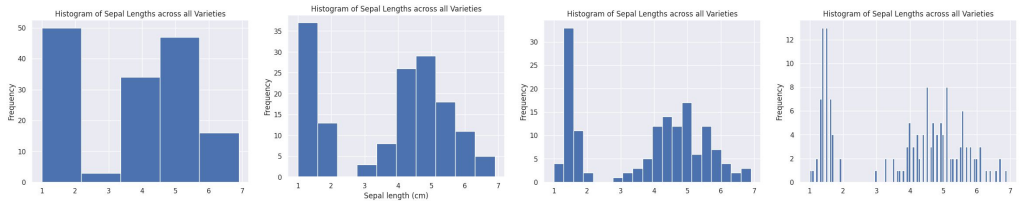
Probability density function (or density) : likelihood of a continuous random variable falling within a particular range of values. It provides a probability distribution for the possible outcomes of a continuous random variable.
The probability that the variable falls within a specific interval [a, b] is given by the area under the curve of the PDF between a and b. Mathematically, this is expressed as the integral of the PDF over the interval [a, b]:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

- Non-negative for all possible values of the random variable
- The total area under the PDF curve over the entire range of values is equal to 1

3. Creating Basic Visualizations

- Probability density estimation
  - Estimate, based on observed data, of an unobservable underlying probability density function
  - Commonly used method: Kernel density estimation (KDE) *sns.kdeplot()*
  - Independent from bins number (histogram)
  - More data points, better estimation



3. Creating Basic Visualizations

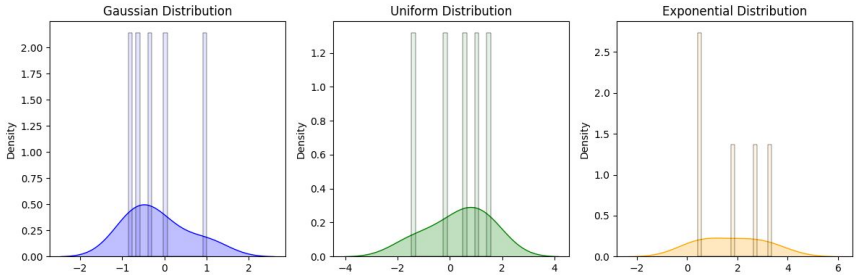
$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \le x \le b, \\ 0 & \text{for } x < a \text{ or } x > b. \end{cases}$$

- Probability density estimation

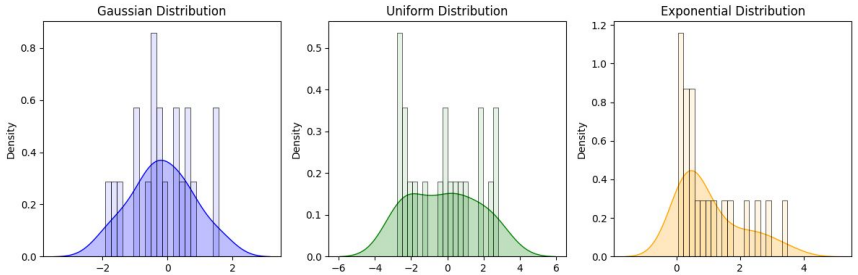
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} .$$

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x} & x \ge 0, \\ 0 & x < 0. \end{cases}$$

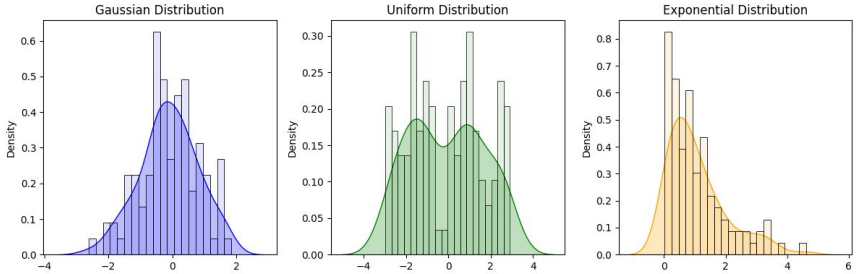
n=5



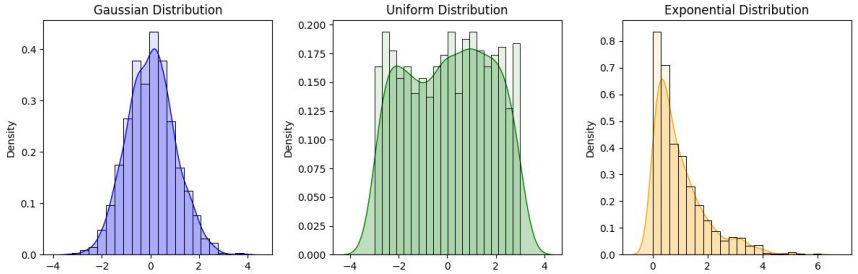
n=20



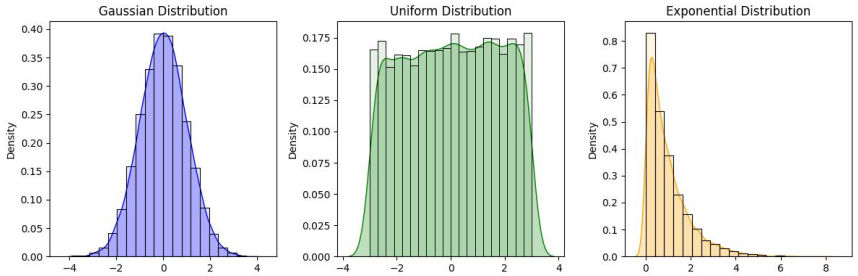
n=100



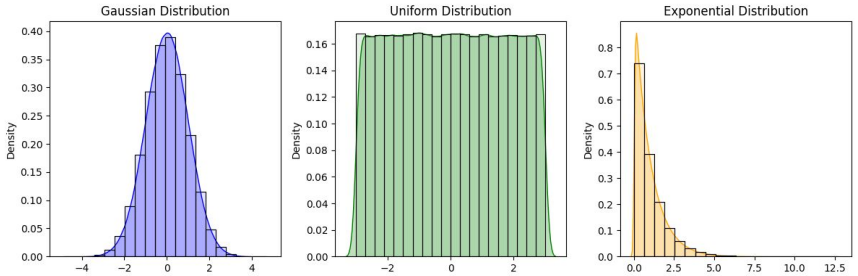
n=1000



n=10000



n=1M

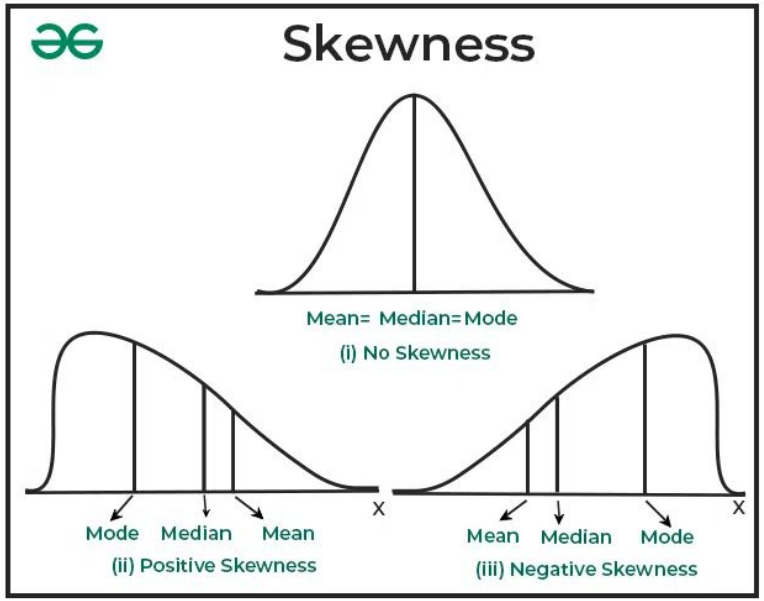


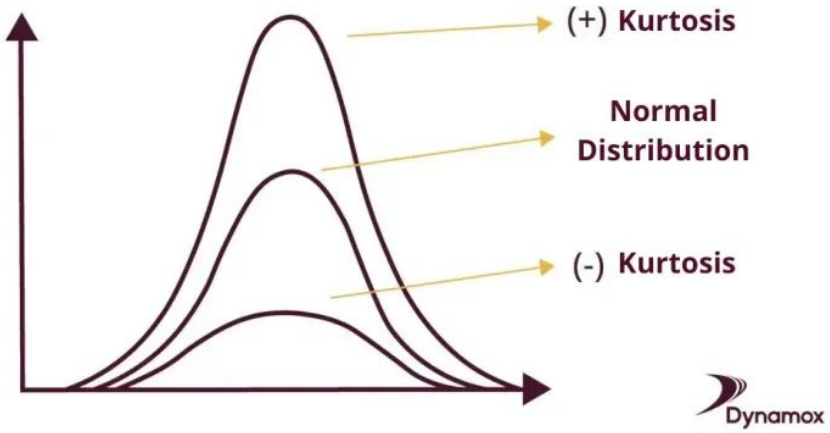
3. Creating Basic Visualizations

- Probability density estimation
  - Skewness
  - Kurtosis

3. Creating Basic Visualizations

- Probability density estimation
  - Skewness: measure of non-symmetry
    - (Difficult) exam results ?
    - Incomes ?
    - Human life ?
    - Gaussian ?
  - Kurtosis: measure of tailedness
    - Outliers ?
    - Uniform distribution ?



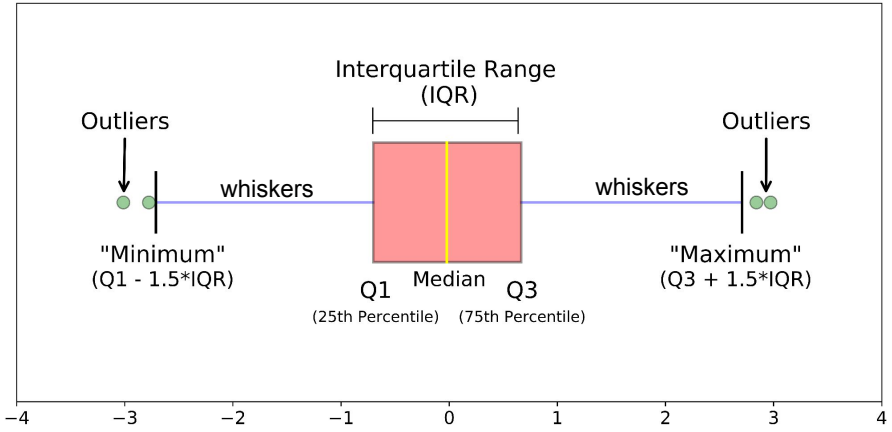


3. Creating Basic Visualizations

- **Box Plot** (or box-and-whisker plot)
  - Shows summaries of distribution of a feature (spread, skewness, median, outlier, etc.)

“Minimum” (Q0, 0th percentile): lowest point excluding outliers (fence)
“Maximum” (Q4, 100th percentile): highest point excluding outliers (fence)
“Median” (Q2, 50th percentile): middle value in the data set
“First quartile” (Q1, lower quartile): median of the lower half
“Third quartile” (Q3, upper quartile): median of the upper half

Whiskers: lines outside the box with a “T” end
Generally (Q1-1.5 IQR), (Q3+1.5 IQR)



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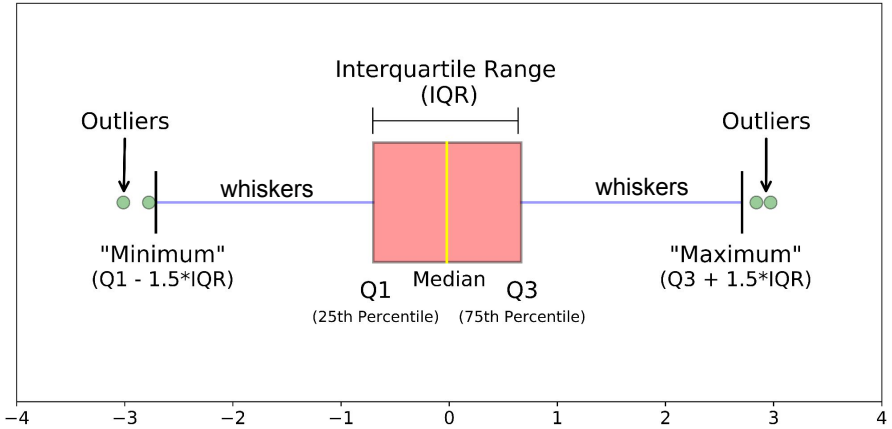
Generally (Q1-1.5 IQR), (Q3+1.5 IQR)

sns.boxplot()
df.boxplot()



3. Creating Basic Visualizations

- Many other plots, feel free to explore (e.g. violin plot, bubble chart, etc.)

Exercises

3.3 Generating scatter plots and histograms

iris.csv dataset again.

a) Plot a single histogram of sepal lengths (all varieties together)

b) Plot 4 histograms (1 per feature), all varieties together. Once with 4 subplots, once overlaid on the same plot.

c) Plot the histograms of petal lengths grouped by variety (3 histograms). Once with 3 subplots, once overlaid on the same plot.

d) Plot the KDE of petal lengths distributions for each variety (overlaid). Compare with the histograms in c)

e) Plot the boxplots, for the 3 varieties, of petal lengths and petal lengths (i.e. 6 boxplots)

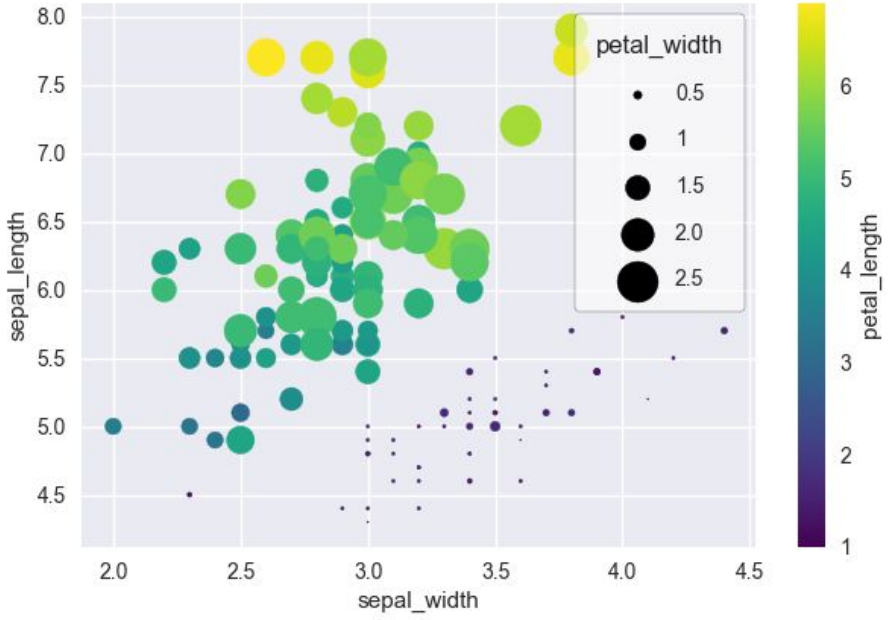
f) Make a scatter plot of sepal length vs petal length. First all same color, then one color per variety

Conclusion: Good for visualization of individual feature, but hard to visualize all categories and all features at once.

4. Visualizing Multivariate Data

4. Visualizing Multivariate Data

- Exploring relationships in multiple dimensions (more than 3 variables)
- Most plots can accommodate 2 or 3 dimensions (e.g. scatterplot + size)
  - More becomes difficult to read
  - Example with 4 variables:

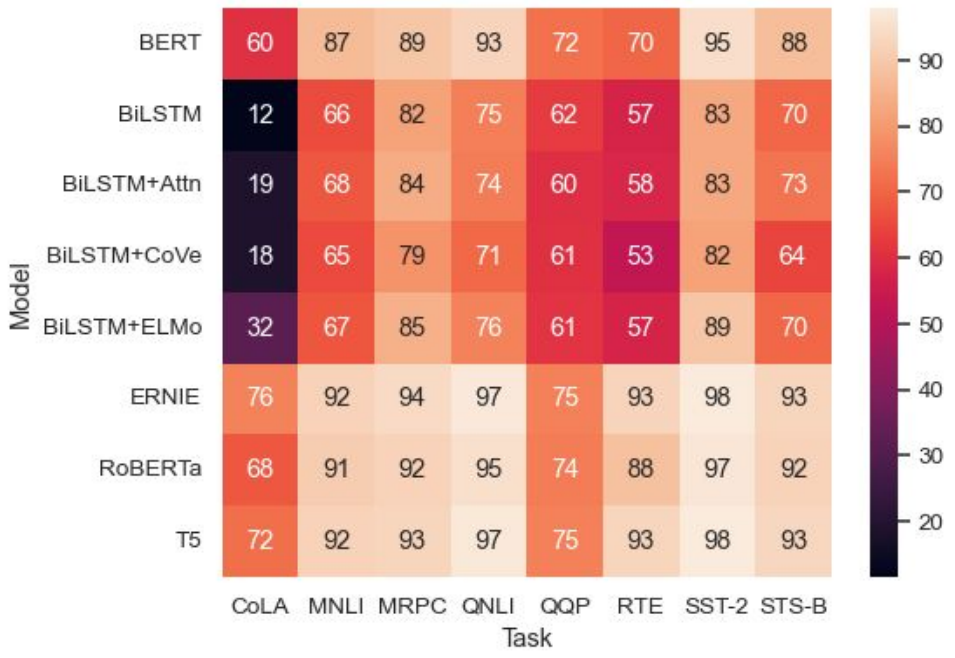


4. Visualizing Multivariate Data

- Heatmaps, Parallel Coordinates, and Scatterplot Matrices

4. Visualizing Multivariate Data

- **Heatmap plot**
  - Plot rectangular data as a color-encoded matrix

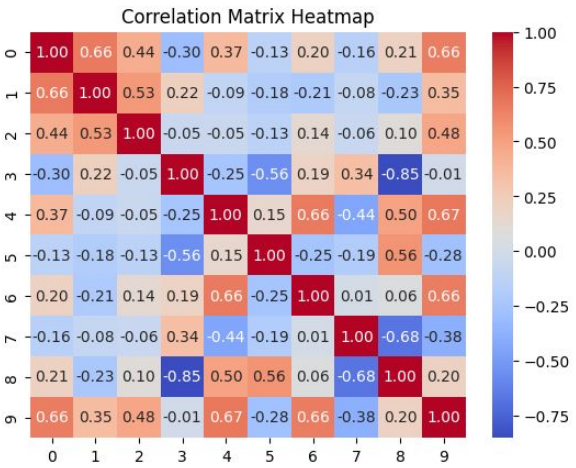


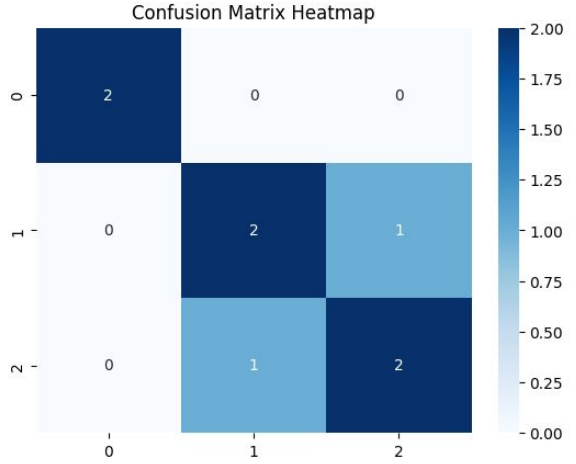
sns.heatmap()

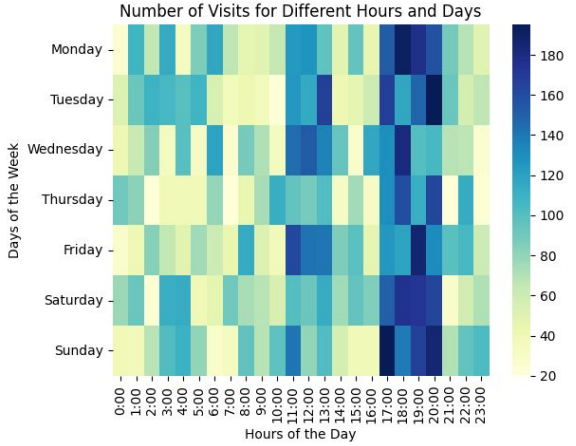
4. Visualizing Multivariate Data

- Heatmap plot
  - Plot rectangular data as a color-encoded matrix
    - Correlation matrix
    - Confusion matrix
    - Temporal patterns in time series data
    - etc.

sns.heatmap()



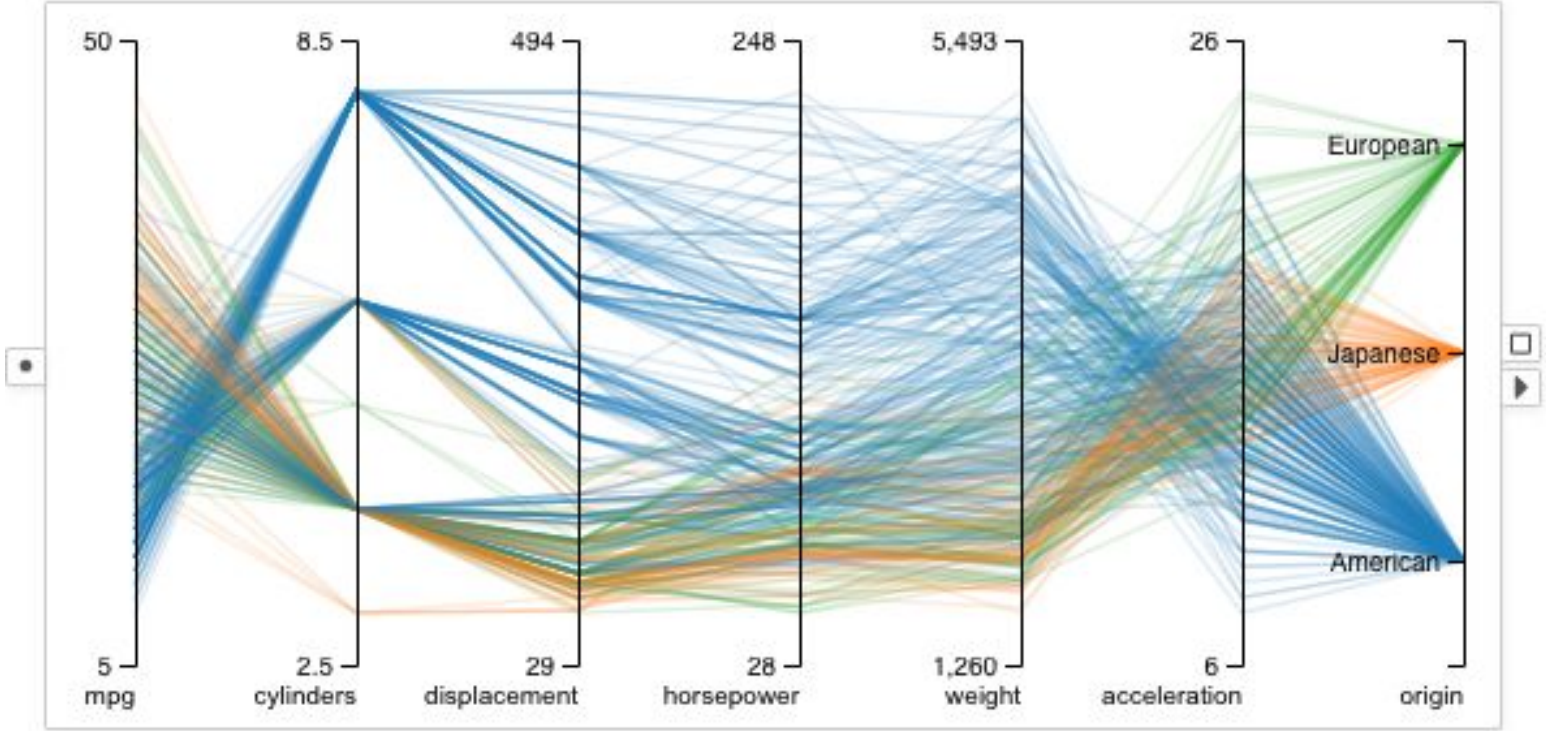




4. Visualizing Multivariate Data

- Parallel Coordinates plot
  - Compare observations across multiple numerical variables
  - Each feature is represented by a separate axis
  - Each observation is plotted horizontally

pd.plotting.parallel_coordinates()



4. Visualizing Multivariate Data

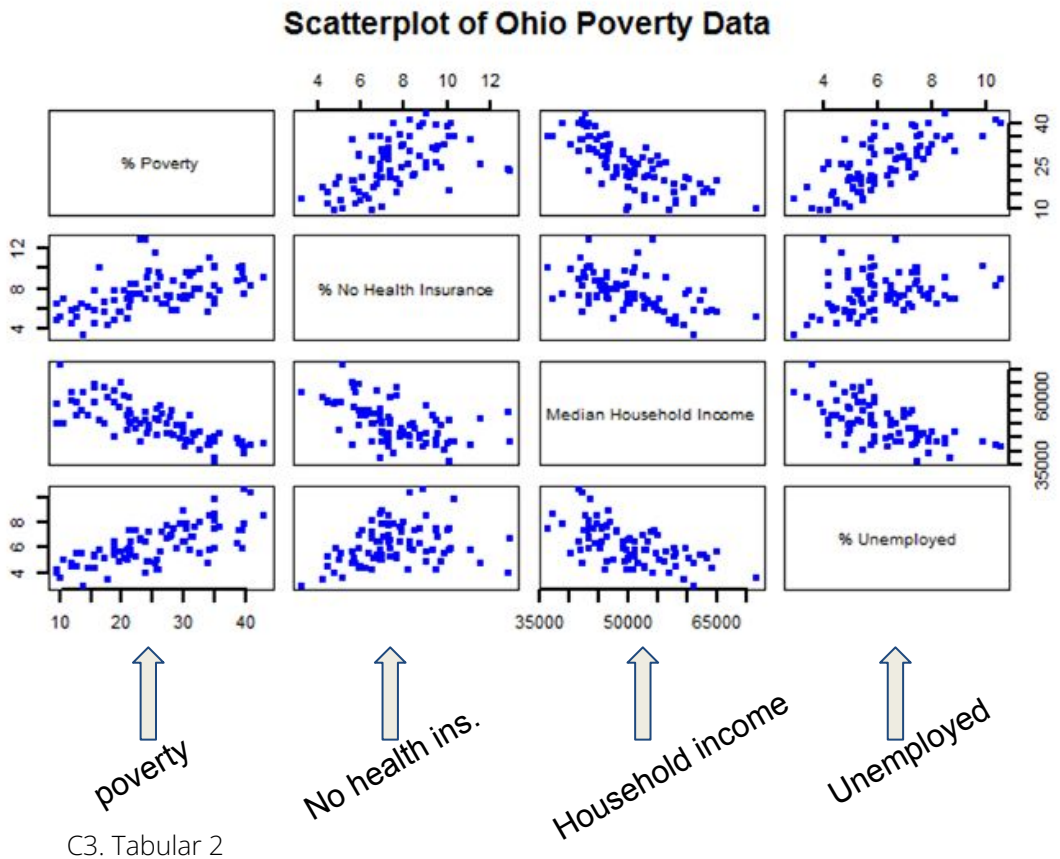
- Scatterplot Matrices
  - Scatter plot created for every pairwise combination of variables
  - Shows relationship (e.g. correlation) between multiple variables

poverty

No health ins.

Household income

Unemployed



sns.pairplot()

Exercises

3.4 Creating a heatmap, parallel coordinates plot and scatterplot matrix

iris.csv dataset again.

a) Make a single plot representing a scatterplot matrix (colored by variety) of the entire dataset with scatter-plots (pair-wise) and density estimations (univariate, in the diagonal). Comment

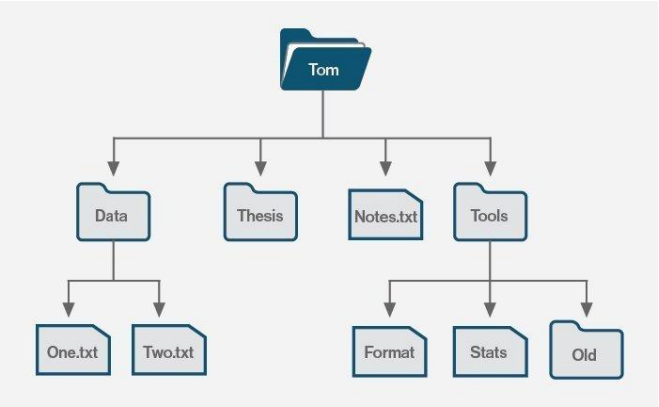
b) Make a parallel coordinates plot. Comment

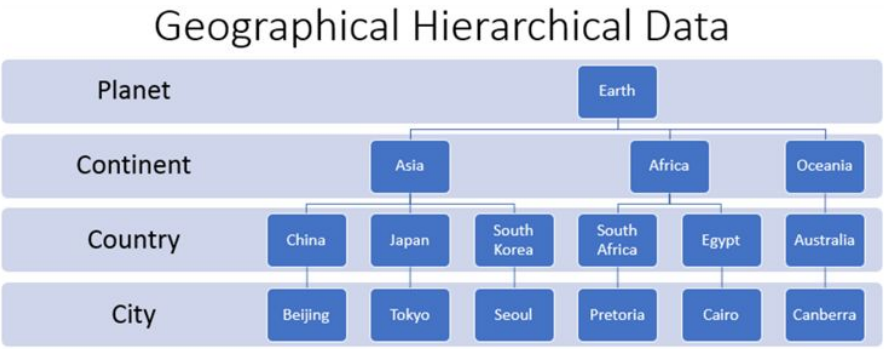
c) Plot a heatmap of correlation of all features. Comment

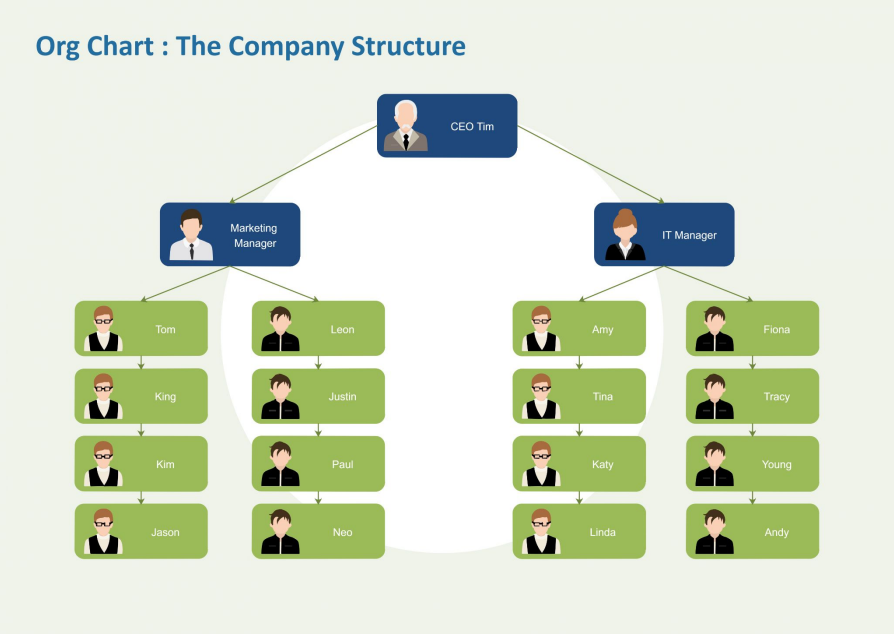
5. Hierarchical Data Visualization

5. Hierarchical Data Visualization

- **Hierarchical data**: Items are linked to each other in parent-child relationships in an overall tree structure
  - Computer file system
  - Geographical data
  - Organization chart (CEO, departments, employees)



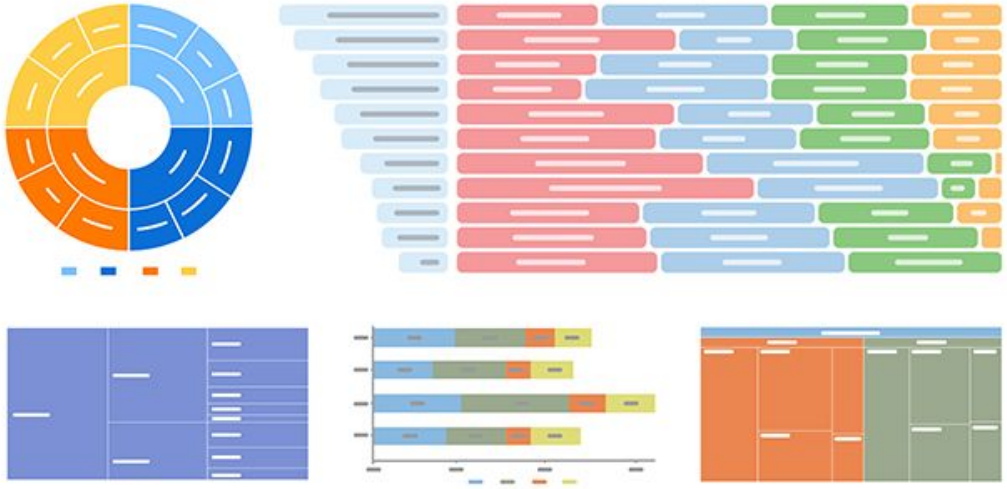




5. Hierarchical Data Visualization

- **Hierarchical data**: Items are linked to each other in parent-child relationships in an overall tree structure
- Our brains are engineered to recognize order and rank in visual content instantly (⭐⭐⭐⭐ vs ⭐⭐)

  → hierarchical data visualizations to create compelling narratives for your audiences



Top 5 Hierarchical Data Visualizations for Data Stories

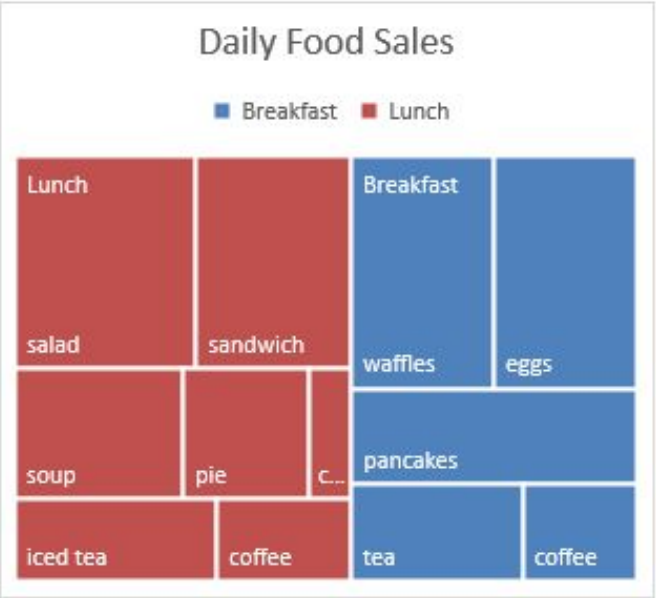
5. Hierarchical Data Visualization

- Treemap Chart, Sunburst Chart
  - Visualization of hierarchical and categorical data

5. Hierarchical Data Visualization

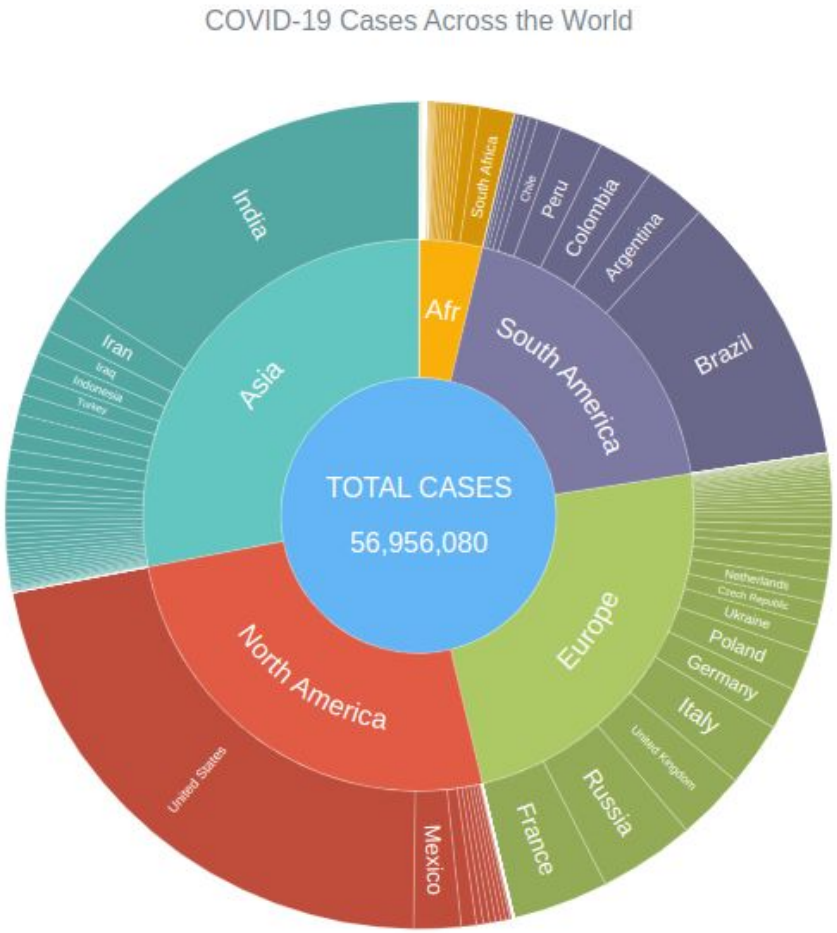
- **Treemap Chart**
  - Used for large, hierarchical data sets
  - Capture (1) The value of data points (2) The structure of the hierarchy
  - Outermost rectangle is the top of the hierarchy

px.treemap()



5. Hierarchical Data Visualization

- **Sunburst Chart**
  - Similar to treemap
  - Each level of the hierarchy is represented by one ring
  - Innermost circle is the top of the hierarchy



px.sunburst()

5. Hierarchical Data Visualization

- Plotly library
  - Express vs graph objects
    - Plotly Express: many functions that can create entire figures at once
    - Graph objects can do more complex plots
    - Plotly Express internally calls Graph objects
  - Can do interactive graphs, we will use that later
    - For non-interactive: *fig.show(renderer="png")*
  - Important parameters to understand:
    - path (hierarchy)
    - values (size)
    - color (...)
    - etc.

	country	continent	year	lifeExp	pop	gdpPercap	iso_alpha	iso_num
0	Afghanistan	Asia	1952	28.801	8425333	779.445314	AFG	4
1	Afghanistan	Asia	1957	30.332	9240934	820.853030	AFG	4
2	Afghanistan	Asia	1962	31.997	10267083	853.100710	AFG	4
3	Afghanistan	Asia	1967	34.020	11537966	836.197138	AFG	4
4	Afghanistan	Asia	1972	36.088	13079460	739.981106	AFG	4
5	Afghanistan	Asia	1977	38.438	14880372	786.113360	AFG	4
6	Afghanistan	Asia	1982	39.854	12881816	978.011439	AFG	4
7	Afghanistan	Asia	1987	40.822	13867957	852.395945	AFG	4
8	Afghanistan	Asia	1992	41.674	16317921	649.341395	AFG	4
9	Afghanistan	Asia	1997	41.763	22227415	635.341351	AFG	4
10	Afghanistan	Asia	2002	42.129	25268405	726.734055	AFG	4
11	Afghanistan	Asia	2007	43.828	31889923	974.580338	AFG	4
12	Albania	Europe	1952	55.230	1282697	1601.056136	ALB	8
13	Albania	Europe	1957	59.280	1476505	1942.284244	ALB	8

Exercises

3.5 Designing a Treemap Chart and a Sunburst Chart

We use the gapminder.csv dataset. Statistics and other information about social, economic and environmental development at local, national and global levels.

With plotly, use *fig.show(renderer="png")* for now, interactive plots will be introduced later.

a) Plot and compare two Treemap charts with a life expectancy color map, and rectangle size representing the population: One in 1952 and one in 2007.

To consider: what color range for comparison? How to set the hierarchy?

b) Plot the Sunburst Chart of gdpPercap in 1952 (color) and population (size). Why is it difficult to see differences with default color range or range_color=[min_gdp,max_gdp]? What solution can be found?

6. Visualizing Temporal Data

6. Visualizing Temporal Data

- Time series data: data that is recorded over consistent intervals of time

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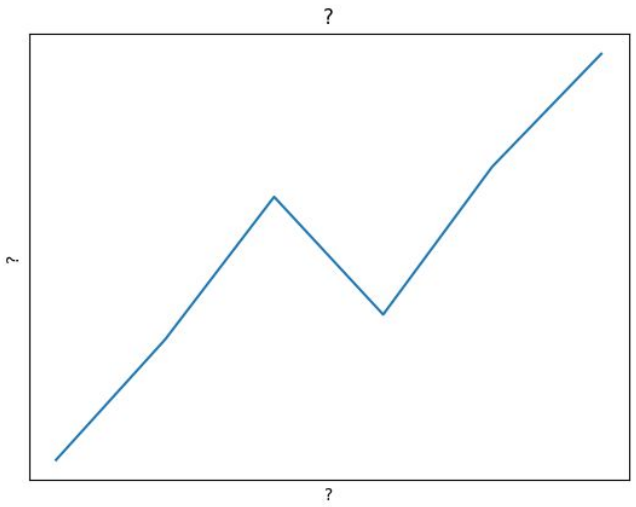
- Time series data: data that are recorded over consistent intervals of time
- Examples:
  - Number of magicpass sold over the years
  - ISC students tiredness during a 4h class
  - Temperature in Sion in the past years



6. Visualizing Temporal Data

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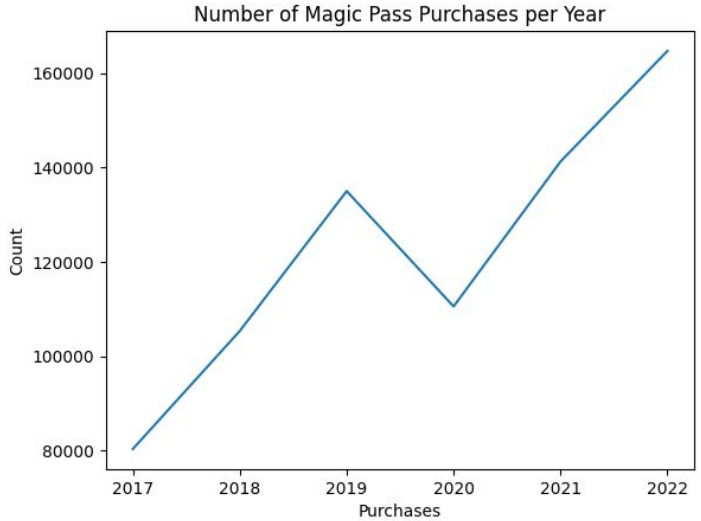


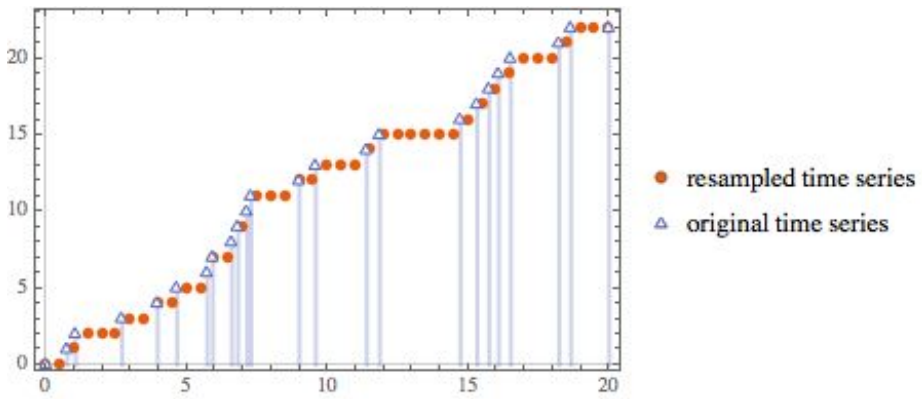
	year	purchases
0	2017	80353
1	2018	105353
2	2019	135000
3	2020	110554
4	2021	141258
5	2022	164700

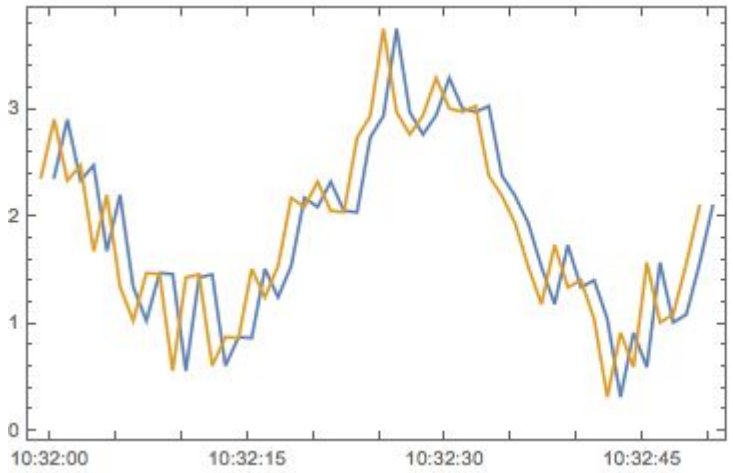
6. Visualizing Temporal Data

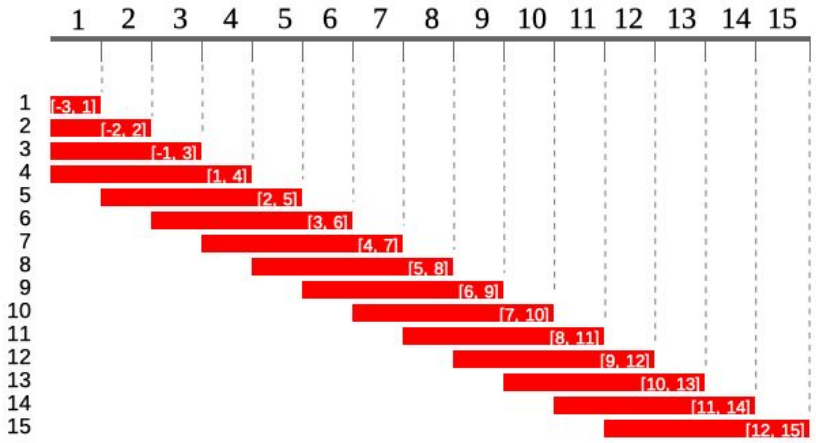
- Time series data: data that are recorded over consistent intervals of time
- Examples:
  - Number of magicpass sold over the years
  - ISC students tiredness during a 4h class
  - Temperature in Sion in the past years
- Pandas for time series data
  https://pandas.pydata.org/docs/user_guide/timeseries.html

6. Visualizing Temporal Data

- Time series operations
  - Resampling: transform data into equispaced data
    - transform data occurring in irregular time intervals
    - change daily data to weekly or monthly (down-sample)
  - Shifting: shift or "lag" the values back and forward in time
    - Find correlation or pattern with past data
  - Rolling-window (or moving):
    - e.g. moving average
    - smooth out short-term fluctuations and highlights long-term trends







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    - e.g. moving average
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```
df.head(10)
```

✓ 0.0s

date	value
2000-01-01	0
2000-01-02	1
2000-01-03	2
2000-01-04	3
2000-01-05	4
2000-01-06	5
2000-01-07	6
2000-01-08	7
2000-01-09	8

```
df.resample('3D').mean()
```

✓ 0.0s

date	value
2000-01-01	1.0
2000-01-04	4.0
2000-01-07	7.0

```
df.shift(periods=3)
```

✓ 0.0s

date	value
2000-01-01	NaN
2000-01-02	NaN
2000-01-03	NaN
2000-01-04	0.0
2000-01-05	1.0
2000-01-06	2.0
2000-01-07	3.0
2000-01-08	4.0
2000-01-09	5.0

```
df.rolling(3).sum()
```

✓ 0.0s

date	value
2000-01-01	NaN
2000-01-02	NaN
2000-01-03	3.0
2000-01-04	6.0
2000-01-05	9.0
2000-01-06	12.0
2000-01-07	15.0
2000-01-08	18.0
2000-01-09	21.0

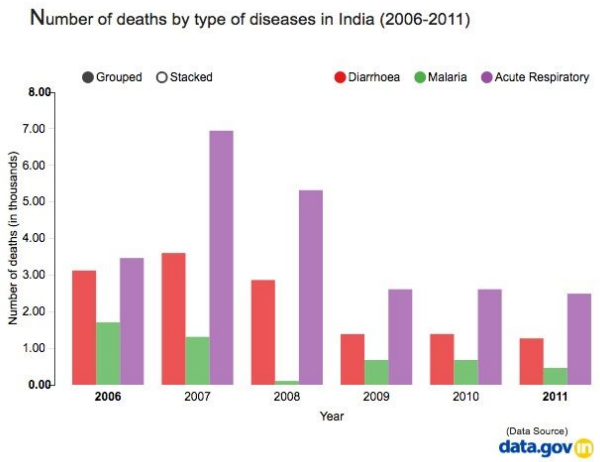
```
df.rolling(3).mean()
```

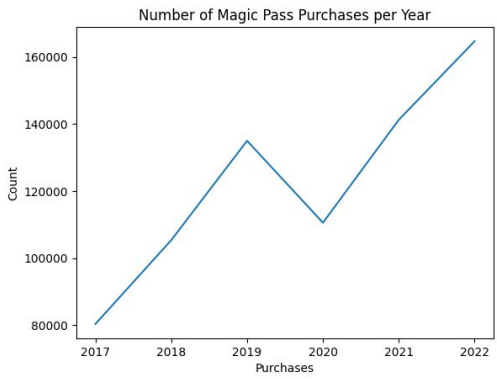
✓ 0.0s

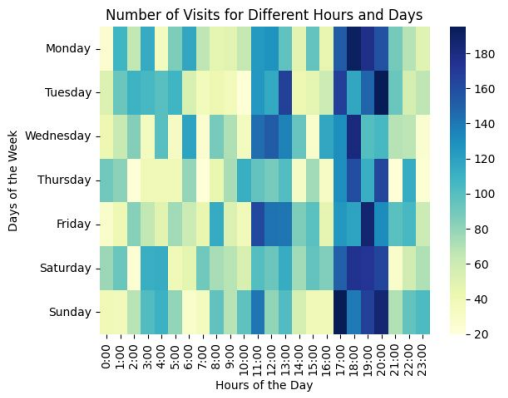
date	value
2000-01-01	NaN
2000-01-02	NaN
2000-01-03	1.0
2000-01-04	2.0
2000-01-05	3.0
2000-01-06	4.0
2000-01-07	5.0
2000-01-08	6.0
2000-01-09	7.0

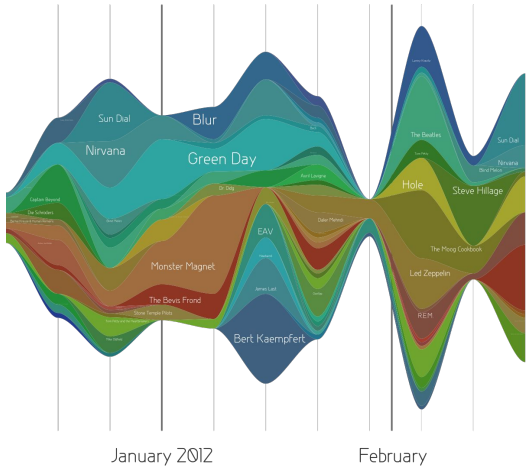
6. Visualizing Temporal Data

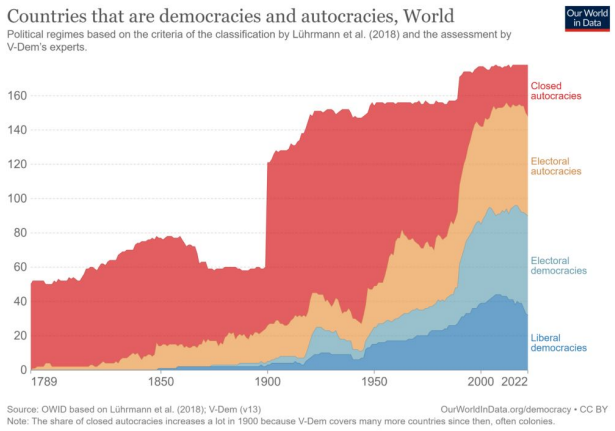
- Time series visualization
  - Bar and Line Plots (Seen in Section 3)
  - Heatmap (Seen in Section 4)
  - Other (e.g. stream graph, stacked area chart, etc.)











6. Visualizing Temporal Data

- **Time series patterns**
  - Seasonality: data experience regular and predictable changes that recur e.g. every calendar year
  - Cycle: rises and falls that are not of a fixed frequency
  - Trend: long-term increase or decrease in the data (Cycle can be mistaken for a trend if looking at a too short time window)

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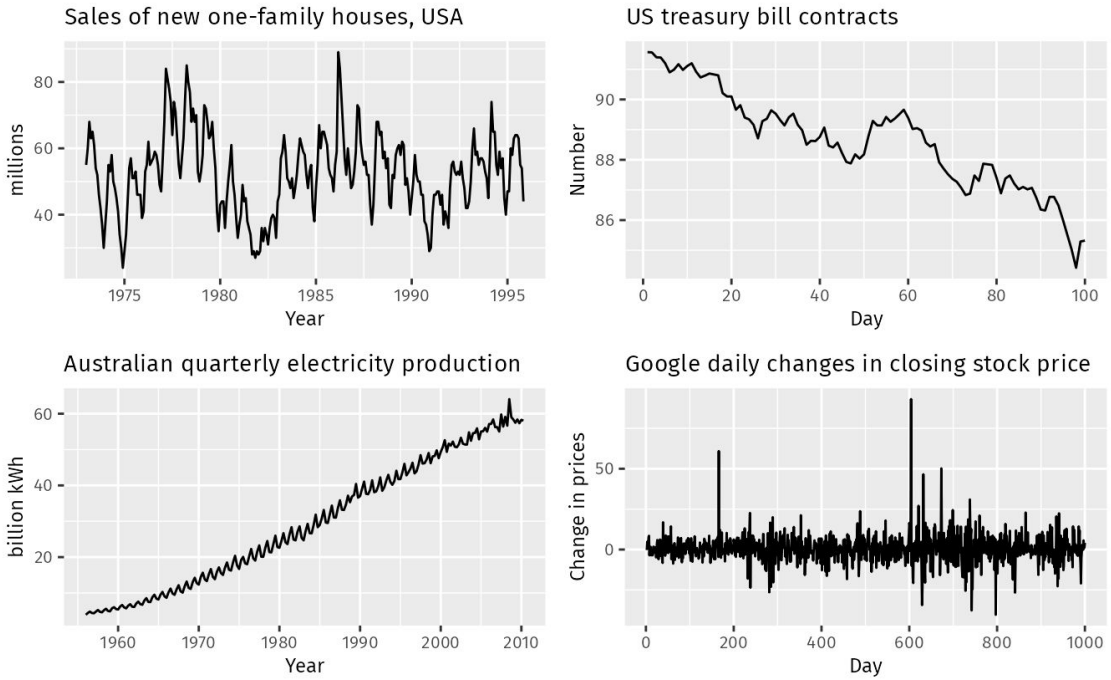

?

6. Visualizing Temporal Data

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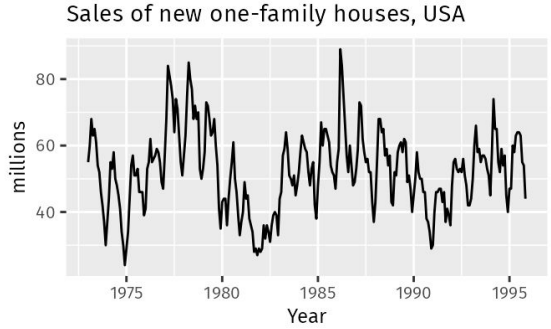
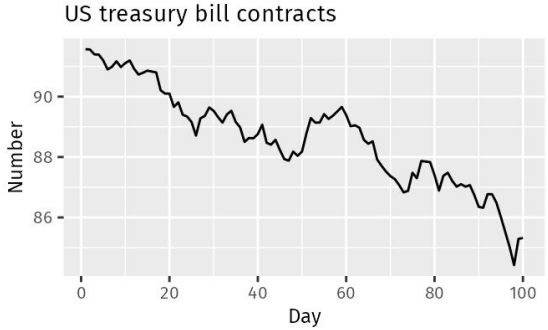
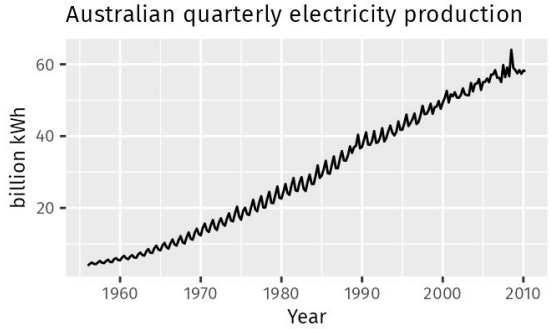
S, C

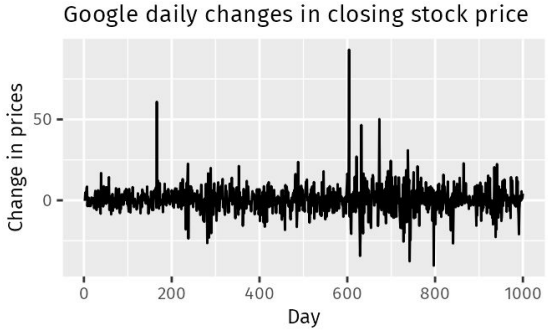




T (could be C?)

T, S





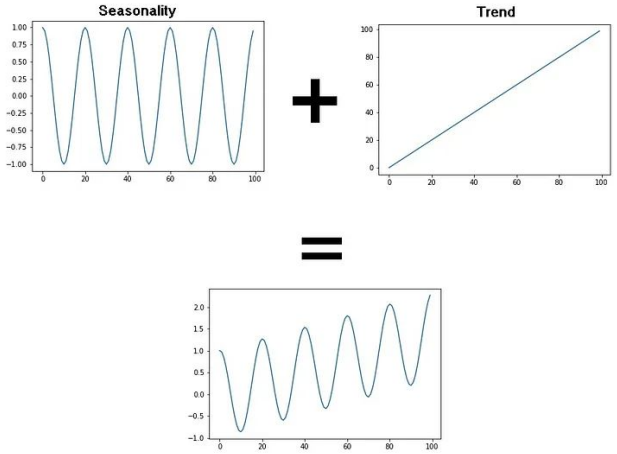
No pattern
(seems random)

6. Visualizing Temporal Data

- Time series patterns
  - Seasonality: data experience regular and predictable changes that recur e.g. every calendar year
  - Cycle: rises and falls that are not of a fixed frequency
  - Trend: long-term increase or decrease in the data (Cycle can be mistaken for a trend if looking at a too short time window)

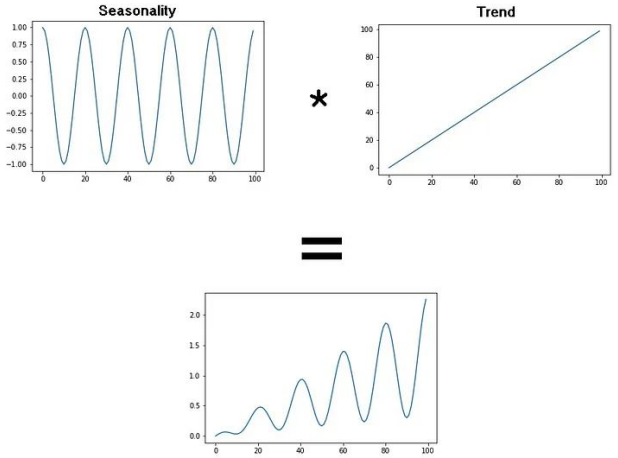
Combination:

additive



Magnitude of the seasonal fluctuations does not vary with the level of time series

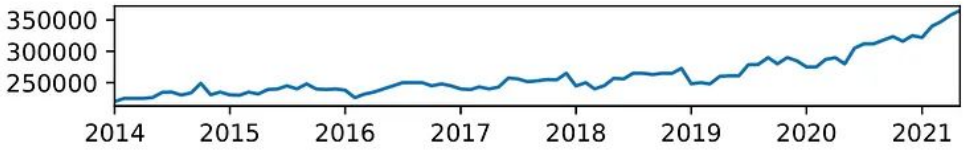
multiplicative



Seasonal fluctuations increase or decrease proportionally with increases and decreases in the level of the series

6. Visualizing Temporal Data

- **Finding time series patterns**
  - Seasonal decomposition (*statsmodels.tsa.seasonal.seasonal_decompose()*)
    - model “additive” or “multiplicative”
    - returns seasonal, trend, and residual



monthly median condo (apartment) prices for the Province of Quebec and the Montreal Metropolitan Area

https://towardsdatascience.com/finding-seasonal-trends-in-time-series-data-with-python-ce10c37aa861

6. Visualizing Temporal Data

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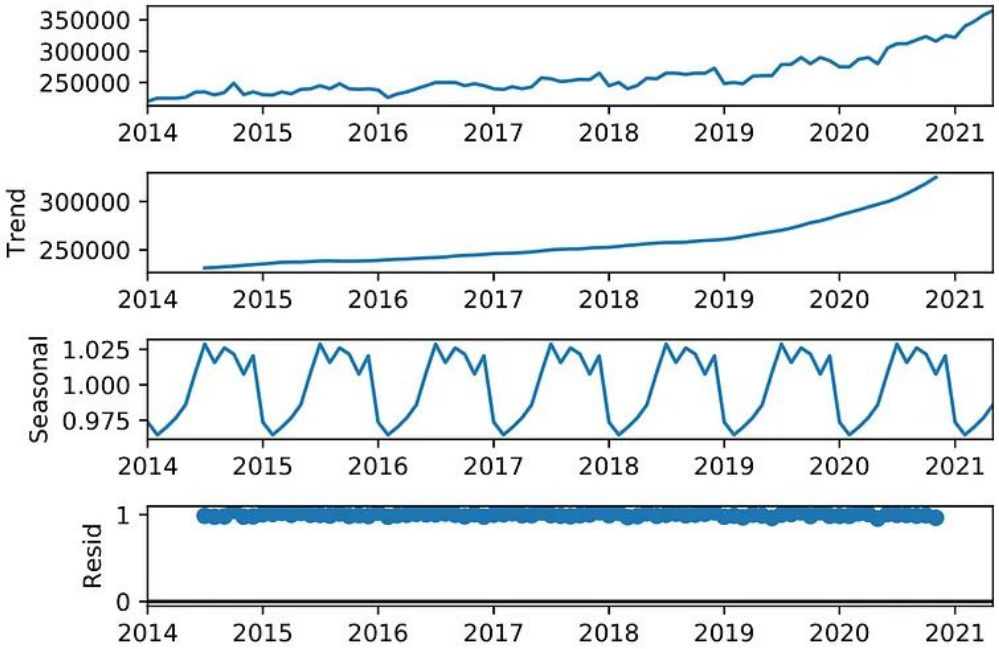


monthly median condo (apartment) prices for the Province of Quebec and the Montreal Metropolitan Area

Multiplicative model

Random noise in the data that isn’t accounted for in the trend or seasonality

https://towardsdatascience.com/finding-seasonal-trends-in-time-series-data-with-python-ce10c37aa861

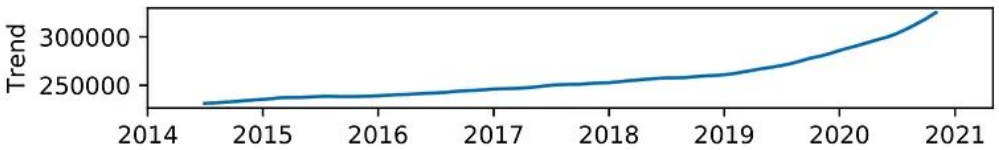
6. Visualizing Temporal Data

- Finding time series patterns
  - Seasonal decomposition (*statsmodels.tsa.seasonal.seasonal_decompose()*)
    - **model “additive” or “multiplicative”: How to choose ?**
      - Amplitude of the seasonality increases or decreases with the trend ?
      - Compute and plot both models, compare the residuals (additive close to 0; multiplicative close to 1)

https://towardsdatascience.com/finding-seasonal-trends-in-time-series-data-with-python-ce10c37aa861

6. Visualizing Temporal Data

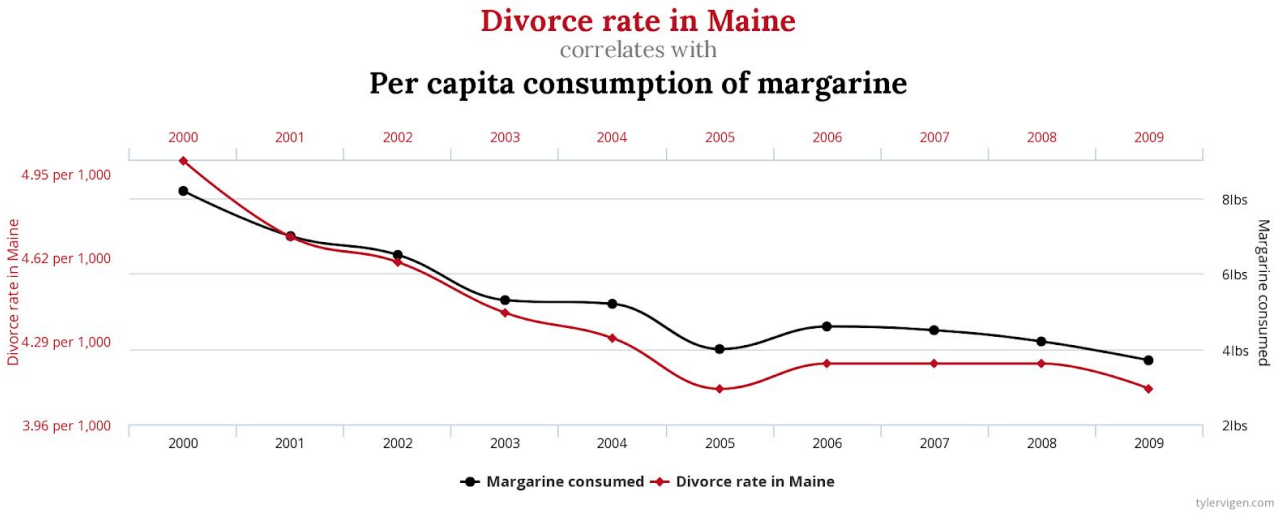
- Finding time series patterns
  - Seasonal decomposition (*statsmodels.tsa.seasonal.seasonal_decompose()*)
    - model “additive” or “multiplicative”
    - returns seasonal, trend, and residual
  - Trends can be correlated with other variables (find correlated trend)
    - e.g. mortgage rates (higher costs for owners),
    - population (more demand)



6. Visualizing Temporal Data

- Finding time series patterns
  - Seasonal decomposition (statsmodels.tsa.seasonal.seasonal_decompose())
    - model “additive” or “multiplicative”
    - returns seasonal, trend, and residual
  - Trends can be correlated with other variables (find correlated trend)

    - Warning: Correlation is not necessarily causation !



sitn.hms.harvard.edu

	datestamp	co2
0	1958-03-29	316.1
1	1958-04-05	317.3
2	1958-04-12	317.6
3	1958-04-19	317.5
4	1958-04-26	316.4
5	1958-05-03	316.9
6	1958-05-10	NaN
7	1958-05-17	317.5
8	1958-05-24	317.9
9	1958-05-31	NaN

Exercises

3.6 Creating a time series plot with a sample dataset

We use the co2_levels.csv dataset. It represents the weekly CO2 levels measured in Mauai, Hawaii between 1958 to 2001.

a) Plot the data resampled weekly, monthly and yearly

b) Plot the yearly (52 weeks) rolling mean of the co2_levels. Add the confidence intervals as +/- stdev

c) Plot the seasonal, trend, and residual. Is the data seasonal? Cyclic? With trend?

Recap

- Effective data visualization
- Basic visualizations (Bar and Line Plots, Crosstab and Stacked bar charts, Scatter Plots, Histograms, probability density)
- Multivariate data (Heatmaps, Parallel Coordinates, and Scatterplot Matrices)
- Hierarchical data (Treemap, Sunburst charts)
- Temporal data
  - Operations: Resampling, shifting and rolling
  - Visualization: Bar plot, line plot, heatmap etc.
  - Time-series patterns: Trend, cycle, seasonality

Questions ?